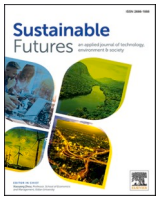
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Balancing relational and contractual governance to foster innovation: Evidence from smes in morocco's textile Sector

Hanane Bougadir [a,*], Antonio García Sánchez [b], Mustapha Zahir [a]

[a] Laboratoire Interdisciplinaire de Recherche en Environnement, Management, Energie et Tourisme (LIREMET), Cadi Ayyad University, Morocco
[b] Grupo de Investigación Análisis Económico y Economía Política, Universidad de Sevilla, Spain

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ABSTRACT

This study investigates the innovation behaviors of subcontracting small and medium-sized enterprises (SMEs) in Morocco's textile industry, focusing on the governance mechanisms influencing these behaviors. SMEs significantly contribute to Morocco's economy but face intensified global competition and complexities within multinational supply networks. Subcontracting emerges as a strategic response, providing SMEs with avenues for growth and innovation. Yet, the specific governance conditions enabling or constraining subcontractor innovation remain inadequately explored. Adopting an exploratory qualitative approach, this multi-case study examines three Moroccan textile SMEs engaged in international subcontracting. Employing thematic analysis of interview data, observations, and secondary sources, the research identifies critical factors influencing innovation behaviors. Findings reveal a nuanced interplay between contractual and relational governance mechanisms. While formal contracts provide structural clarity, relational governance, grounded in trust, proactive communication, and mutual commitment, proves essential for sustained innovative activities. Internal innovation capacities, particularly dedicated funding and skilled personnel, significantly enhance innovation potential but are constrained by limited resources. Moreover, cooperative relationships characterized by mutual learning, risk-sharing, and transparent communication foster innovation more effectively than transactional subcontracting approaches. This research underscores the necessity of balancing relational trust with contractual protection to optimize SMEs' innovative performance. Practical implications include recommendations for SME managers to cultivate long-term, trust-based relationships complemented by strategic contractual agreements. Policymakers are advised to encourage relational governance practices through targeted support programs. Future studies should extend these insights across different sectors and larger samples to validate the findings' broader applicability.

1. Introduction

Small and medium-sized enterprises (SMEs) represent over 90 % of global businesses [1] and are crucial for job creation and economic growth [2,106]. However, they increasingly face challenges related to sustainability and competitiveness due to limited resources, complex market dynamics, and heightened global competition exacerbated by recent economic disruptions [3–5]. The COVID-19 pandemic has further intensified these challenges, forcing SMEs to seek innovative approaches to maintain competitiveness and operational resilience [5,3]. Interorganizational linkages, including subcontracting arrangements, have become essential strategic responses for addressing these mounting pressures [6–8].

Subcontracting has evolved significantly beyond simple task delegation to encompass multiple essential facets of organizational operations, creating complex multilateral networks that require sophisticated governance approaches [7,10]. This practice represents a fundamental aspect of effective organizational functioning, with subcontractors increasingly recognized as key strategic partners rather than mere service providers [9]. The growth of subcontracting arrangements has been particularly notable in recent years (2020–2025), driven by digital transformation, supply chain restructuring post-pandemic, and the need for specialized capabilities in rapidly evolving markets [10,8].

However, this expansion introduces considerable complexity in governance mechanisms. While formal agreements establish roles and responsibilities, recent research emphasizes that relational governance

* Corresponding author.
E-mail addresses: Bougadirha@gmail.com (H. Bougadir), acichez@us.es (A.G. Sánchez), Zahirmustapha@gmail.com (M. Zahir).

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approaches, characterized by open communication, mutual benefits, and trust-based collaboration, provide significant advantages, including increased information sharing, supply chain flexibility, and enhanced innovation capacity [11,12]. Nonetheless, purely cooperative approaches can create reluctance to share critical knowledge due to potential resource leakage and intellectual property concerns, particularly relevant in today's knowledge-intensive economy [13]. Therefore, successful subcontracting relationship management requires a nuanced understanding of both contractual and relational mechanisms to balance trust-based collaboration with necessary legal protections.

Recent literature has increasingly focused on innovation as a critical outcome of effective subcontracting relationships [13,55,67]. Innovation thrives on trust, receptiveness to new ideas, and calculated risk-taking [24–27], yet the specific governance conditions that most effectively support innovation within SMEs operating as subcontractors remain inadequately explored. Current studies predominantly examine either contractual or relational governance in isolation, with limited investigation into their interactive effects on innovation behaviors [17–19].

Despite growing recognition of subcontracting's potential to foster innovation, critical gaps persist in the existing literature. First, the literature remains fragmented regarding the mechanisms by which governance structures influence subcontractor innovation behaviors. While studies acknowledge that subcontracting can positively correlate with innovation [20,108,22,23], the specific governance conditions that enable or constrain these innovation behaviors remain unclear, particularly in emerging economy contexts where institutional frameworks may differ significantly from developed markets. Second, there is insufficient theoretical integration of how contractual and relational governance mechanisms interact dynamically to influence innovation outcomes. Most existing research treats these governance forms as substitutes rather than examining their potential complementary effects [77,78], leaving a significant theoretical void in understanding optimal governance configurations for innovation. Third, limited empirical evidence exists regarding how SMEs in developing economies, particularly in traditional industries like textiles, navigate the complex interplay between governance mechanisms and innovation pressures within international subcontracting networks [109]. This gap is particularly significant given the increasing integration of emerging market SMEs into global value chains [58,59].

To address these gaps, this study investigates the following primary research question: How does the interplay of contractual and relational governance mechanisms within subcontracting relationships influence SME innovation behaviors? This primary question is supported by examining what internal factors drive SME innovation behaviors in subcontracting relationships, how external cooperation mechanisms shape innovation outcomes in SME subcontracting, and under what conditions contractual and relational governance mechanisms complement or substitute each other in fostering innovation.

This study offers several novel contributions to the intersection of governance theory, innovation management, and SME literature. Theoretically, this research uniquely applies an extended Theory of Planned Behavior (TPB) framework to examine governance-innovation relationships in subcontracting contexts, providing a fresh theoretical lens that integrates behavioral intention theories with governance mechanisms. This approach fills a critical theoretical gap by explaining not just what governance mechanisms influence innovation, but why and how they shape behavioral intentions toward innovation. Empirically, unlike previous studies that focus on single governance forms or developed economy contexts, this research provides the first systematic examination of governance mechanism complementarity in fostering SME innovation within Morocco's textile industry, a traditional sector undergoing digital transformation pressures. This emerging economy perspective offers crucial insights for understanding governance-innovation dynamics in resource-constrained environments. Methodologically, the study employs a rigorous multiple-case study approach with methodological triangulation, combining in-depth interviews, non-participant observation, and secondary data analysis. This methodological rigor, enhanced by the application of the PRISMA framework for case selection transparency, provides robust empirical foundations for theory building in underexplored contexts. Practically, the research generates actionable insights for SME managers navigating complex governance decisions in subcontracting relationships, offering evidence-based guidance on balancing relational trust with contractual protection to optimize innovative performance, a critical capability for post-pandemic recovery and future resilience.

This study proceeds as follows: First, we explore the fundamental concepts associated with innovation behaviors and the role of governance mechanisms in facilitating innovative behavior, grounding our analysis in recent theoretical developments. Second, we present our research methodology, emphasizing the rigor and transparency of our case selection process. Third, we analyze the data to consolidate empirical findings and discern significant patterns through systematic thematic analysis. Finally, we outline the main conclusions, theoretical implications, and practical recommendations of our findings, contributing to both scholarly knowledge and managerial practice in SME innovation governance.

2. Literature review

2.1. Theoretical foundations

2.1.1. Theory of planned behavior

Proposed by Ajzen [28] as an extension of the theory of reasoned action [29], the theory of planned behavior (TPB) posits that an individual's intention to perform a behavior is determined by attitudes (positive or negative evaluations), subjective norms (perceived social pressure), and perceived behavioral control (ease of performing the behavior) [30].

The TPB discusses circumstances where individuals are not entirely in charge of their behavior. For example, studies in various countries (e.g., Germany, China, Pakistan, and Malaysia [31]; Italy [32]; Spain [33]) demonstrate the influence of the TPB variables on firms' technology adoption and the positive effects of personal attitude and perceived behavioral control on innovations. Furthermore, Ma et al. [34] observe some firm owners from China and bear out the positive impacts of personal attitude, subjective norms, and perceived behavioral control on innovative investments of firm owners.

Behavioral beliefs reflect the anticipated consequences of a specific behavior [35]. Subjective norms represent an individual's perception of social pressure from important referents concerning whether to perform the behavior [36]. Perceived behavioral control, encompassing self-efficacy in executing an action, predicts behavioral intentions and subsequent actions [37]. Executives' attitudes towards innovation, stakeholders' environmental innovation norms, and the firm's perceived behavioral control (including collaborative innovation, resource availability, and R&D activities) positively influence long-term firm innovation behavior and performance [38].

Bano and Siddiqui [39] investigate Indian individuals' adoption of smart technologies such as the Internet of Things (IoT), artificial intelligence (AI), virtual reality systems, and augmented reality systems and find a positive influence of personal attitude and subjective norm on the adoption of these technologies. Previous research establishes a positive correlation between subjective norms and perceived usefulness [40], as well as between perceived behavioral control and the intention to utilize the information system [41,42]. Personal attitude also helps enterprises reduce barriers to adopting technologies, and therefore, it improves firm executives' abilities to find innovative solutions [36].

Given the nature of entrepreneurial work, which involves discovering and exploiting opportunities, finding creative solutions, and managing limited resources, innovative skills and a strong sense of self-efficacy in these areas are crucial [43]. This self-efficacy, encompassing

the ability to identify and leverage opportunities, devise creative solutions, and operate under constraints, influences an entrepreneur's choices and engagement in innovative activities [44]. Furthermore, entrepreneurial attitudes focused on identifying and creating new opportunities enhance firms' competitive advantages and innovation performance [45].

Despite the well-established predictive power of the theory of planned behavior for intention and behavior, its extensive application has led to scholarly examination [11]. It's important to note that beyond the core TPB variables, other factors can significantly influence behaviors [46,47]. For instance, past behavior should be considered as a significant variable in certain contexts, while in the field related to integrity, moral hazard, or moral concepts are critical factors [11].

2.1.2. Resource-based view theory

The Resource-Based View (RBV), pioneered by [124] and refined by [112], posits that firms achieve competitive advantage through valuable, rare, inimitable, and non-substitutable (VRIN) resources. In the context of SME innovation, RBV provides crucial insights into how resource constraints and capabilities influence innovation behaviors within subcontracting relationships.

RBV is particularly relevant for understanding SME innovation because it explains how firms with limited resources can leverage external partnerships to access capabilities they lack internally. Recent studies demonstrate that SMEs often compensate for resource limitations through strategic alliances and subcontracting arrangements that provide access to complementary resources [118,113]. This perspective is especially pertinent for understanding how Moroccan textile SMEs navigate resource scarcity while pursuing innovation within international subcontracting networks.

The theory helps explain why some SMEs excel in innovation despite resource constraints, while others struggle. Internal innovation capacities, such as dedicated R&D personnel, innovation funds, and absorptive capacity, represent firm-specific resources that, when combined with external governance mechanisms, can create sustainable competitive advantages [119]. For SMEs operating as subcontractors, the challenge lies in developing and maintaining these internal capabilities while simultaneously managing external dependencies.

Contemporary RBV research emphasizes the dynamic nature of resources and capabilities, particularly in rapidly changing environments [122,123]. This dynamic perspective is crucial for understanding how SMEs must continuously reconfigure their resource portfolios to maintain innovation capacity within evolving subcontracting relationships. The theory also illuminates how governance mechanisms can either enhance or constrain SMEs' ability to access and utilize external resources for innovation purposes.

2.2. Innovation behaviors

Innovative behavior (IB) is understood as the degree to which innovations are actually implemented within various firm divisions [48,49,105,106]. It refers to all individual actions that are aimed at the formation, introduction, and application of new and useful ideas to improve the organization [50,51]. Employees are typically the initial identifiers of workplace issues, which enables them to propose new ideas [52]. Consequently, innovative behavior among employees is considered a fundamental element that enhances firm performance [53] and establishes a basis for bottom-up innovation [54].

Recent studies confirm that innovative behavior effectiveness depends significantly on organizational context and external support mechanisms [115,116]. Specifically, [3] demonstrates that SMEs in emerging economies face distinct innovation challenges compared to their developed economy counterparts, including limited access to technology, restricted financial resources, and institutional voids. This research highlights the importance of understanding innovation behaviors within specific economic and cultural contexts, particularly for SMEs operating in traditional industries undergoing digital transformation. Post-pandemic research reveals that SMEs increasingly rely on digital innovation platforms and virtual collaboration tools to maintain innovation momentum despite resource constraints [121,117,110]. This shift has profound implications for how governance mechanisms operate in virtual subcontracting environments, requiring new approaches to trust-building and knowledge sharing.

Recognizing the need for sustainable competitive advantage, firms are increasingly leveraging various management, organizational, and business configuration approaches to promote innovation [55–57], a phenomenon supported by extant literature identifying a wide range of individual, organizational, and environmental factors of innovative activity [58–60]. Driven by various triggers such as new challenges, inconsistencies, or discontinuities in professional practices [61,62], innovative behaviors encompass a wide array of activities, including problem-solving, fulfilling customer demands, and altering work patterns [51].

Beyond simplistic typologies like incremental versus radical innovation, investigating the sources of innovation reveals complex managerial situations. Innovative behavior is a multistage process [63]. The initial stage, idea generation, commences when employees identify opportunities or issues. The creative potential of an individual is enhanced when their skills, abilities, and mental state align with available environmental resources, increasing their likelihood of creative accomplishments [64].

During the second stage, seeking support, employees rely on supervisors or colleagues for their new ideas. Trust is crucial for fostering social exchanges [65]; its absence leads to self-protective behaviors hindering innovation [65]. Furthermore, inter-firm cooperation can balance the exploration of new knowledge with the exploitation of existing knowledge [28]. Large enterprises encounter external barriers; however, SMEs confront more restrictive internal barriers, frequently lacking the diverse and specialized knowledge and skills essential for promoting innovation and reaching target markets [66,67].

In the final stage, idea implementation, the generated idea becomes a concrete result through testing and the launch of new products/services or processes. Within subcontracting, optimal collaboration occurs when subcontractors actively participate beyond basic production in prototype refinement, contributing to iterative product design and enhanced efficiency [68,21]. Erik Eriksson et al. [21] demonstrate that partnering and procurement approaches significantly influence subcontractor involvement in innovation activities, with collaborative procurement methods fostering greater innovation engagement than traditional approaches. Bocquet's research [20] confirms the positive influence of collaborative agreements on a subcontractor's likelihood of innovation.

2.3. Subcontracting relationships, governance, and innovation behaviors

Given the recognized significance of the general contractor-subcontractor relationship in collaborative projects, both industry and scholarly communities have increasingly acknowledged that adversarial interactions can negatively impact project cost, schedule, and quality, and increase the likelihood of disputes, claims, and litigation [11]. Consequently, an increasing number of experts and researchers are investigating the collaboration between main contractors and subcontractors [69,70].

Recent advances in governance theory emphasize the critical role of hybrid governance structures that combine formal and informal mechanisms [78]. [114] provide evidence that purely contractual or purely relational approaches are insufficient in today's complex supply networks, advocating for dynamic governance configurations that adapt to project requirements and relationship maturity

Emerging research on digital governance platforms shows how technology mediates traditional governance mechanisms, creating new opportunities for transparency and trust-building while introducing novel coordination challenges [71,111]. These technological

developments are reshaping how subcontracting relationships are managed and how innovation occurs within these partnerships.

The subcontract delineates the rights, obligations, and responsibilities of both the general contractor and the subcontractor, along with the mechanism for risk-sharing [71,72]. In addition, the subcontractor executes a specific segment of the work on behalf of the general contractor. The latter is responsible for managing subcontractors and ensuring compliance with client specifications, as well as coordinating activities among a group of subcontractors. To some extent, the general contractor and various subcontractors exhibit interdependence [73,74].

This interdependence, however, is often accompanied by power dynamics where control can be exerted through submission, socialization, and indoctrination [75]. Relational governance, encompassing the enforcement of commitments, obligations, promises, activity expectations, and shared objectives [76], is crucial for establishing cooperative inter-firm relationships [14,15]. Research indicates that reliance solely on a formal contract, although important, is insufficient for fostering these relationships [16,77,78].

Instead, various interface factors, such as low levels of conflict, mutual trust, effective coordination, and open communication, significantly influence the relationship throughout the process [79–82]. Moreover, relational governance necessitates the development of social identity and trust [78], which subsequently fosters innovative behavior [83,84]. Critically, the ordering party's failure to uphold this tacit agreement within a short timeframe can severely erode the subcontractor's trust [85].

General contractors typically establish three types of relationships with their subcontractors: long-term business relationships, short-term business relationships, and hostile-dependent relationships [107]. The collaboration between a general contractor and a subcontractor yields favorable performance outcomes advantageous to both entities [86]. To be precise, if they work together as long-term partners, both parties stand to achieve [87]. Long-term partnerships enhance performance and substantially improve project delivery [88].

Long-term cooperation intention denotes the behavioral orientation or ideological inclination of both parties to engage in a sustained collaborative effort [11]. The subcontractor expects the ordering party to regularly propose new services, while the ordering party expects the subcontractor to prioritize it in case of competition with another ordering party and to refrain from directly contacting the ordering party's client.

Furthermore, many general contractors maintain one-time or short-term business relationships with subcontractors possessing the specific capabilities or qualifications to perform specialized work to a high standard [89]. Asymmetric power dynamics and conflicting interests may precipitate specific modalities of interfirm cooperation. Nevertheless, subject to a positive assessment of performance and collaborative efficacy, the general contractor will demonstrate a sustained intent to engage in future cooperative endeavors [90].

2.4. *Conceptual framework development*

Building on the theoretical foundations of TPB and RBV, this study proposes a conceptual framework that integrates internal innovation capacities, governance mechanisms, and external cooperation factors as determinants of SME innovation behaviors in subcontracting relationships.

Fig. 1, Fig. 2.

The conceptual framework presents a comprehensive view of SME innovation behaviors in subcontracting relationships by synthesizing insights from the Theory of Planned Behavior (TPB) and the Resource-Based View (RBV). It identifies three key, interrelated dimensions that collectively influence innovation outcomes: internal innovation

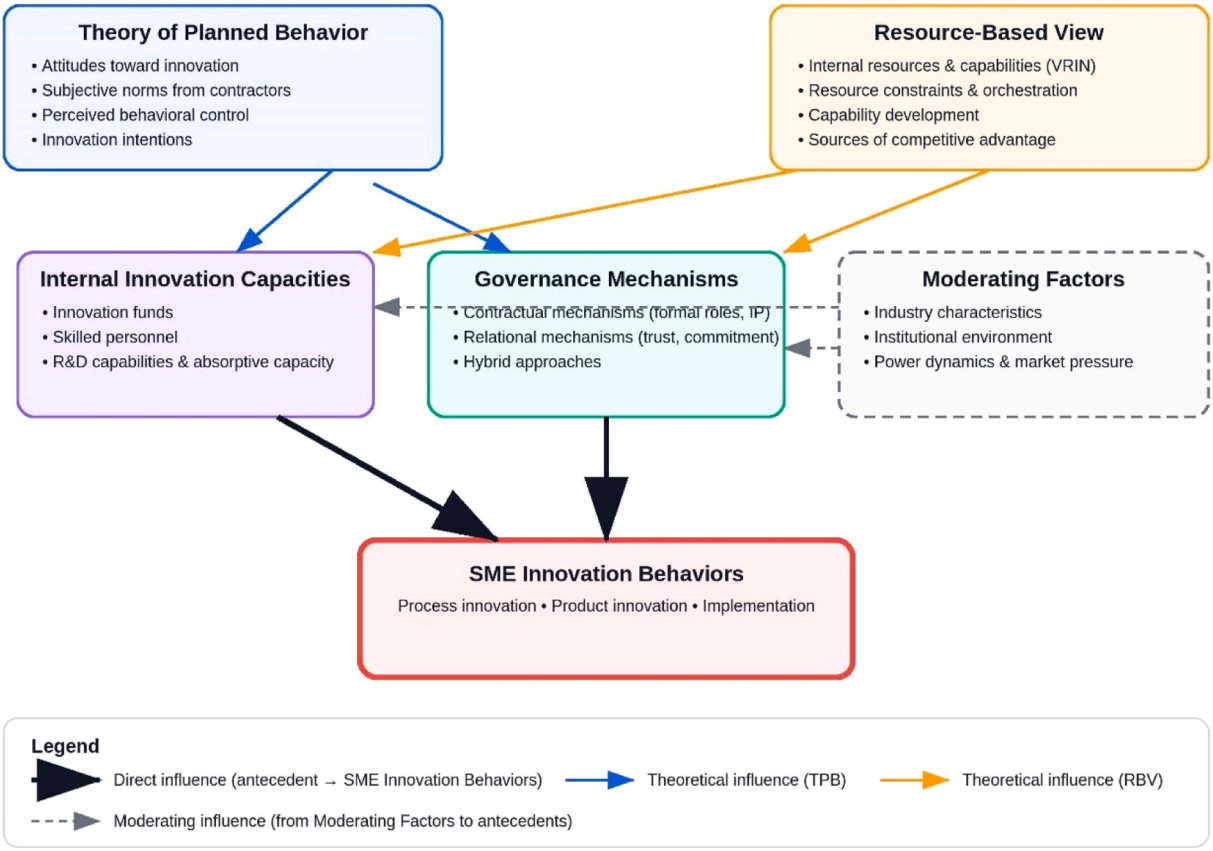


Fig. 1. Conceptual framework of SME innovation behaviors in subcontracting relationships.

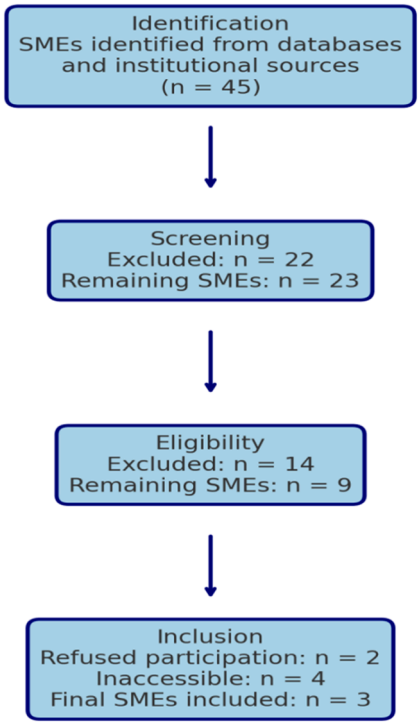


Fig. 2. PRISMA diagram illustrating SME selection process.

capacities, governance mechanisms, and external cooperation factors. These dimensions are not isolated but function as a dynamic system shaped by moderating contextual elements.

The first dimension, Internal Innovation Capacities, is grounded in the RBV perspective, which emphasizes the strategic importance of firm-specific resources. These include financial assets such as innovation funds, skilled human resources, and advanced R&D capabilities. In addition, organizational competencies, such as absorptive capacity and knowledge integration systems, form the backbone of innovation potential. From the RBV standpoint, these elements represent valuable, rare, inimitable, and non-substitutable (VRIN) resources that equip SMEs to engage in innovation, even under resource-constrained conditions.

The second dimension, Governance Mechanisms, draws from both transaction cost economics and relational governance theories. It comprises two complementary forms: contractual governance, which involves formal agreements, legal structures, and clearly delineated roles; and relational governance, which emphasizes trust, mutual commitment, and open communication. The framework suggests that effective governance requires a flexible equilibrium between these mechanisms, responsive to the maturity of the partnership and the complexity of the innovation tasks involved.

The third dimension, External Cooperation Factors, is influenced by the TPB framework and highlights the behavioral and relational context shaping SME innovation. This includes subjective norms, expectations from contractors regarding innovation, as well as SMEs' perceived behavioral control and attitudes toward collaborative innovation. Beyond psychological determinants, this dimension also incorporates practical aspects such as partnerships for R&D, consultant collaborations, and mechanisms for sharing knowledge, all of which enhance innovation capacity through external engagement.

Crucially, the framework acknowledges the role of moderating factors that shape the interplay among these dimensions. These include sector-specific characteristics (traditional versus tech-driven industries), institutional influences such as regulatory support and infrastructure, power imbalances within subcontracting networks, and the market-driven pressures that compel firms to innovate. These contextual variables can either facilitate or hinder the effectiveness of innovation strategies.

The framework further stresses the dynamic interdependence among its core dimensions. For example, robust internal capabilities can enhance an SME's ability to engage in external collaborations, while well-functioning governance structures can strengthen the synergy between internal and external innovation drivers. External cooperation, in turn, may compensate for internal deficiencies by offering access to complementary skills and resources.

Overall, the theoretical synthesis of TPB and RBV provides a nuanced lens for understanding SME innovation behavior. TPB accounts for the psychological and social underpinnings of innovation decisions, while RBV focuses on the structural and resource-related enablers. When considered together, these perspectives highlight that successful innovation in subcontracting contexts depends on both the intention and ability to innovate, supported by strategic governance.

3. Research methodology

This study employs an exploratory qualitative methodology, adopting a multiple-case study approach to examine innovation behaviors and governance mechanisms in Moroccan textile SMEs engaged in international subcontracting relationships. The methodology is designed to provide in-depth insights into complex, context-specific phenomena rather than statistical generalization.

3.1. Research design and approach

This study explores the conception and evolution of innovative behavior in SMEs involved in subcontracting relationships with major corporations. We opted for an exploratory qualitative methodology to delve deeply into understanding and elucidating a complex situation [91]. According to Fossey, Harvey, McDermott, and Davidson [92], qualitative research aims to address questions concerned with developing an understanding of the meaning and experience dimensions of humans' lives and social worlds [93]. posits that we should view qualitative research as a formal, systematic process that fundamentally aims to discover answers to issues by applying scientific procedures that facilitate the acquisition of new knowledge.

We chose the case study method for this qualitative research because this method is the most appropriate for describing, explaining, predicting, and controlling processes inherent to various phenomena, whether individual or collective [94]. To mitigate the risk of overly focusing on a single case or producing less generalizable results, we adopted a multiple-case study, which, according to Miles [95], strengthens the validity of the results. This approach allows for cross-case comparison and pattern identification while maintaining the depth of analysis characteristic of qualitative inquiry.

3.2. Inclusion and exclusion criteria

The selection of SMEs for this multiple-case study followed clearly defined inclusion and exclusion criteria to ensure relevance, consistency, and depth in the research findings. These criteria were established prior to the selection process to maintain objectivity and transparency in the case selection.

3.2.1. Inclusion criteria

SMEs were selected based on several stringent inclusion criteria that ensured their relevance to our research objectives. First, companies had to operate specifically within the Moroccan textile and clothing industry, aligning with our focus on this traditional yet evolving sector. Second, enterprises needed to employ between 10 and 200 permanent employees, conforming to the Moroccan national classification criteria for SMEs. Third, selected companies had to demonstrate established long-term international subcontracting relationships, defined as

ongoing partnerships lasting at least three consecutive years, ensuring sufficient experience and relationship maturity for meaningful analysis.

In addition, companies were required to show explicit engagement in innovation activities, whether product or process innovation, within the last three years (2020–2023), providing recent and relevant innovation experiences. Geographic location within Morocco's main textile production regions, including Casablanca-Settat, Rabat-Salé-Kénitra, or Tangier-Tetouan-Al Hoceima regions, ensured accessibility and representation of key industrial areas. Companies also needed to demonstrate willingness and availability to participate in the research, including access to key personnel for interviews, and the ability to provide secondary data sources such as company reports, innovation records, and partnership agreements.

3.2.2. Exclusion criteria

Enterprises were systematically excluded from the study if they failed to meet the inclusion criteria or presented characteristics that would compromise the study's focus and coherence. Companies with fewer than 10 or >200 permanent employees were excluded to maintain consistency with SME definitions. Those without established international subcontracting relationships or with relationships lasting less than three years were excluded, as insufficient relationship experience would limit insights into governance mechanisms and innovation dynamics.

Businesses failing to demonstrate innovation activities within the defined period (2020–2023) were excluded, as innovation behavior constituted the primary phenomenon of interest. Organizations operating outside the textile and clothing industry were excluded to maintain sectoral consistency and comparability. Companies unwilling to participate or unable to provide access to necessary personnel and documentation were excluded for practical reasons. Finally, enterprises established less than five years ago were excluded to ensure sufficient operational history for meaningful analysis of innovation patterns and relationship development.

3.3. PRISMA framework for case selection

To transparently illustrate the selection process of small and medium-sized enterprises (SMEs) for the multiple-case study, we adopted the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework. Originally developed for systematic reviews in the medical and social sciences, PRISMA has proven to be a valuable tool for ensuring methodological transparency and rigor in case selection processes.

3.3.1. PRISMA selection process

The PRISMA selection process followed four sequential phases that systematically narrowed our initial pool to the final sample. The identification phase began with an extensive search across multiple databases and sources, yielding 127 potential SMEs. This initial pool was compiled through Morocco's Ministry of Industry and Trade SME registry, industry association membership lists from the Moroccan Association of Textile and Clothing Industries (AMITH), regional chamber of commerce databases, and academic and industry network referrals.

The screening phase involved applying basic inclusion criteria related to industry sector, company size, and geographic location, while removing duplicates and inactive companies. This phase also included an initial contact assessment for participation willingness. During this phase, 82 companies were excluded for various reasons: 34 due to size mismatch, 28 for industry mismatch, 12 for geographic constraints, and 8 for being inactive or closed, leaving 45 companies for further evaluation.

The eligibility assessment phase involved in-depth evaluation of the remaining companies' subcontracting relationships, duration, and international scope, assessment of innovation activities through preliminary interviews and documentation review, and evaluation of access to key personnel and documentation availability. This rigorous assessment led to the exclusion of 33 additional companies: 18 for insufficient subcontracting experience, 9 for limited innovation evidence, and 6 for access restrictions, resulting in 12 companies deemed eligible.

The final inclusion phase involved a comprehensive suitability assessment for diversity and representativeness, final confirmation of participation willingness and ethical consent, and resource availability assessment for detailed case analysis. Nine companies were excluded in this final phase: 4 due to resource constraints, 3 for insufficient diversity contribution, and 2 for participant withdrawal, ultimately resulting in the selection of 3 SMEs that formed our final sample.

3.4. Participant recruitment and selection

The recruitment process followed a purposive sampling approach, guided by theoretical considerations including the relevance of firms' subcontracting practices to our research question, diversity of subcontracting relationships across different countries, and varying degrees of innovation activities. This strategic approach ensured that selected cases would provide rich, relevant data for addressing our research objectives.

3.4.1. Recruitment strategy

Our recruitment strategy employed multiple channels and stakeholder engagement to identify and approach potential participants. Initial contact was established through formal introductory letters sent to company leadership explaining research objectives, followed by telephone calls to clarify participation requirements and benefits. Face-to-face meetings with key decision-makers were arranged to establish rapport and trust, during which research information sheets and consent forms were provided.

The recruitment process was enhanced through collaboration with the Moroccan Association of Textile and Clothing Industries (AMITH) for endorsement, engagement with regional development agencies for introductions, and utilization of academic institution partnerships for credibility enhancement. This multi-pronged approach helped overcome initial hesitation and established legitimacy for the research endeavor.

3.4.2. Participant selection within cases

Within each selected SME, participants were recruited based on their roles and knowledge relevance to the research questions. Primary participants included company owners or founders who provided strategic decision-making perspectives, general managers offering operational oversight and innovation implementation insights, technical directors contributing innovation development and technical aspects knowledge, production managers sharing process innovation and operational efficiency experiences, quality managers providing quality improvement and customer satisfaction perspectives, and R&D personnel, where available, offering innovation development and external collaboration insights.

Individual participants were selected based on specific criteria, ensuring their ability to contribute meaningful insights. Each participant was required to have a minimum of two years of experience in their current role, direct involvement in innovation activities or subcontracting relationship management, willingness to participate in extended interviews lasting 60–90 min, and availability for potential follow-up discussions to clarify or expand upon initial responses.

3.5. Ethical considerations

This research adhered to rigorous ethical standards throughout all phases of the study, ensuring participant protection and data integrity while maintaining the highest standards of research conduct.

3.5.1. Ethics approval

The study complies with international ethical standards for research

involving human participants and adheres to the principles outlined in the Declaration of Helsinki. The approval process involved a comprehensive review of research procedures, participant protection measures, data handling protocols, and potential risks and benefits.

3.5.2. Informed consent

All participants provided written informed consent before data collection commenced, following a comprehensive consent process that ensured full understanding of their involvement. The consent process included a clear explanation of research objectives and procedures, information about data collection methods and expected time commitment, assurance of voluntary participation and right to withdraw without penalty, confidentiality and anonymity guarantees, explanation of data storage and disposal procedures, and provision of contact information for the research team and ethics board.

3.5.3. Confidentiality and anonymity

Comprehensive measures were implemented to protect participant and company confidentiality throughout the research process. All company names are anonymized using case numbers (Case 1, Case 2, Case 3) in all documentation and publications. Individual participants are identified only by role designation (e.g., "Technical Director, Case 2″) without personal identifiers. Sensitive commercial information is aggregated or generalized to prevent competitive disadvantage. Data files are stored securely with password protection, and access is restricted to research team members. Audio recordings are transcribed by the research team and deleted after verification and analysis completion.

3.5.4. Data security

Robust data security measures ensure the protection of all collected information throughout the research lifecycle. All data is stored on encrypted, password-protected devices with regular backup procedures. Physical documents are secured in locked filing cabinets within restricted access areas. Access to data is limited strictly to research team members who have signed confidentiality agreements. A data retention period of five years post-publication has been established, after which all data will be securely deleted according to institutional protocols.

3.6. Data collection

Qualitative research does not aim to quantify or measure events but instead seeks to collect descriptive data through the researcher's direct and interactive engagement with the situation under study. The researcher endeavors to understand the phenomenon by considering the perspectives of the people involved and then interpreting these phenomena [99,100]. Our data collection strategy employed multiple sources to ensure a comprehensive understanding and methodological triangulation.

3.6.1. Data sources

Following Yin's [100] identification of potential sources of evidence for case studies, we utilized interviews, observations, documents and records, and physical artifacts. These sources are particularly appropriate for an in-depth investigation of case studies, as they facilitate the collection of rich data and the execution of detailed analyses while providing multiple perspectives on the same phenomena.

3.6.2. Primary data collection

Semi-structured interviews formed the primary data collection method, with interview guides addressed to owners, leaders, directors (financial, technical, and production), and managers (quality, research, and development) of each company. This approach ensured comprehensive data collection across different organizational levels and functional areas. Face-to-face interviews took place in the respondents' places of work because, according to Gelo, Braakmann, & Benetka [101], qualitative data collection works best in natural environments where participants feel comfortable and can reference their daily experiences.

Each interview lasted approximately 90 min and was conducted in the participant's workplace to maintain a natural environment conducive to open discussion. Interviews were conducted in Arabic, French, or English based on participant preference to ensure comfort and accuracy of communication. Digital audio recording was employed with participant consent, supplemented by concurrent observational notes during interviews to capture non-verbal cues and contextual information.

We conducted the interviews during September and October 2023, with most sessions becoming privileged moments that exceeded the initially planned one-hour duration. Despite occasional appointment rescheduling due to operational demands, the interviewees seemed delighted to share their experiences and their specific contributions to the innovations established by their respective companies. This enthusiasm contributed to the richness and depth of the data collected.

Non-participant observation served as an integral complement to interviews, providing insights into daily operations, production processes, and employee interactions, particularly those relating to the practical implementation of innovation. These observations offered contextual depth and complemented the perspectives captured through interviews [103,104]. Structured observation sessions lasted 2–3 h per company visit, focusing on production processes, innovation implementation, and team interactions. Detailed field notes and process mapping were maintained, with observations conducted during normal working hours to capture routine operations and authentic workplace dynamics.

3.6.3. Secondary data collection

Secondary data collection encompassed company reports, industry publications, and market analyses, providing an understanding of broader industry trends and particular innovative practices observed in the studied SMEs. Secondary data sources included company annual reports and financial statements, innovation project documentation and patents, subcontracting agreements and contracts with confidential sections redacted, industry reports from AMITH and the Ministry of Industry, market analysis reports and competitor assessments, and training records and employee development programs. This comprehensive secondary data collection provided triangulation opportunities and contextual background for primary data interpretation.

3.7. Sample size justification

The selection of three SMEs for this study reflects the exploratory nature of qualitative research and aligns with established practices in case study methodology. While recognizing that this sample size represents a limitation regarding the generalizability of findings, this exploratory qualitative research aims to deeply analyze and interpret complex, context-specific phenomena rather than to generalize across a larger population. Following Yin's [98] recommendations on multiple-case study design, our focus emphasizes gaining in-depth insights rather than statistical generalization.

According to Yin [96], sampling in a multiple-case study follows a replication logic rather than a statistical sampling logic, where the guiding question becomes "Which cases are most likely to address our research question?" The recommended number of cases depends on how many are necessary to discover what must be known [97]. These three SMEs were deliberately selected because of their rich experiences with international subcontracting and recent innovation activities, providing valuable illustrative examples for our exploratory study. The sample size aligns with established practices in qualitative case study research, where depth of analysis takes precedence over breadth of coverage, enabling detailed examination of complex phenomena within their natural contexts.

3.8. Data analysis and interpretation

We employed a thematic analysis approach, as outlined by Braun and Clarke [102], to analyze the qualitative data collected through interviews, observational notes, and secondary documentation. This method enabled us to systematically identify, organize, and interpret recurring patterns and relationships within the data, providing a comprehensive understanding of SMEs' innovation behaviors within their specific contexts. The analysis was designed to move beyond descriptive accounts toward theoretically grounded insights that illuminate governance-innovation dynamics in subcontracting relationships.

3.8.1. Analytical framework and approach

Our analytical approach was grounded in the theoretical foundations established in our literature review, particularly the Theory of Planned Behavior and Resource-Based View perspectives on innovation and governance. However, we maintained openness to emergent themes that might challenge or extend these theoretical foundations, adopting an abductive reasoning approach that moved iteratively between data and theory.

The analysis process was designed to address our research questions systematically while allowing for unexpected insights to emerge from the rich qualitative data. We prioritized analytical rigor through systematic procedures, multiple validation approaches, and transparent documentation of analytical decisions throughout the process.

3.8.2. Data preparation and management

Prior to formal analysis, all interview recordings were transcribed verbatim in their original languages (Arabic, French, or English) and then translated into English where necessary. Translation accuracy was verified through back-translation procedures conducted by independent bilingual researchers to ensure meaning preservation and cultural nuance retention.

Observational field notes were typed and formatted consistently, with clear temporal and contextual markers to enable systematic analysis. Secondary documents were digitized and organized thematically to facilitate cross-referencing with primary data sources. All data sources were imported into NVivo software for systematic organization and analysis.

Data anonymization procedures were implemented consistently, with all company names, individual identifiers, and sensitive commercial information removed or coded to protect participant confidentiality while maintaining analytical utility.

3.8.3. Systematic coding process

3.8.3.1. Initial familiarization phase. The analysis process began with thorough familiarization, where the research team immersed themselves in the data through repeated readings of interview transcripts, field notes, and industry documents. This familiarization process involved at least three complete readings of all data sources, with preliminary notes and observations documented throughout.

During familiarization, we paid particular attention to language nuances, cultural references, and context-specific meanings that might influence interpretation. This phase established a foundational understanding of each case's unique characteristics while identifying potential patterns across cases.

3.8.3.2. Initial coding procedures. Initial coding followed familiarization, utilizing NVivo software to conduct inductive, line-by-line coding where descriptive labels were assigned to meaningful data segments. Codes were derived directly from participant statements and field observations, maintaining a close connection to the empirical data while beginning to identify conceptual patterns.

The initial coding process revealed diverse perspectives on innovation drivers, constraints, and governance preferences across the three cases. This diversity provided rich material for subsequent thematic development while highlighting the complexity of governance-innovation relationships in SME contexts.

Coding consistency was maintained through regular team discussions and inter-coder reliability checks. Initial codes were assigned independently by two researchers, with discrepancies resolved through discussion and data re-examination until consensus was achieved.

3.8.3.3. Code development and refinement. Following initial coding, codes were reviewed, refined, and consolidated to eliminate redundancy while preserving meaningful distinctions. Related codes were grouped into code families, and preliminary patterns were identified through systematic comparison across cases and data sources.

Code definitions were developed and documented to ensure consistent application throughout the analysis process. Regular code review sessions were conducted to maintain definitional clarity and prevent analytical drift as new data was incorporated.

3.8.4. Theme development process

3.8.4.1. From codes to themes. Theme development involved grouping related codes into broader thematic clusters based on shared patterns and meanings. This process moved beyond simple aggregation toward identifying underlying conceptual relationships that connected codes at higher levels of abstraction.

Table 1, Table 2, Table 3, Table 4, Table 5, Table 6, Table 7, Table 8, Table 9.

Theme development was guided by theoretical sensitivity while remaining grounded in empirical data. Each theme was required to demonstrate coherence within itself, distinctiveness from other themes, and adequate empirical support across multiple data sources.

3.8.4.2. Theme refinement and definition. Theme refinement involved iterative review processes to ensure coherence, consistency, and theoretical significance. Each theme was clearly defined to reflect its conceptual boundaries and empirical foundations:

Internal Innovation Capacities: Encompasses the influence of internal resources (financial assets, human capital, organizational capabilities) and constraints on SMEs' innovation behaviors, reflecting Resource-Based View perspectives on competitive advantage through resource configuration and utilization.

External Cooperation and Innovation: Captures how collaboration with external actors (contractors, consultants, research institutions) shapes innovation outcomes through knowledge sharing, resource access, and collaborative problem-solving, extending open innovation principles to SME subcontracting contexts.

Governance Mechanisms and Innovation Dynamics: Explores the

Table 1
Examples of the initial coding process.

Transcript Extract	Initial Code	Analytical Notes
"We created an internal innovation fund to boost ideas."	Internal funding for innovation	Dedicated resource allocation strategy
"Consultants occasionally overlook our inputs."	External collaboration challenges	Power dynamics in partnerships
"Contracts often fail to protect us adequately."	Contractual protection limitations	Governance mechanism insufficiency
"Trust-based relationships foster creativity"	Trust-driven innovation	Relational governance enabler
"Lack of financial capacity limits innovation."	Resource constraints on innovation	RBV theoretical connection
"Long-term relations are better than short-term contracts."	Relational governance preference	Governance mechanism choice

Table 2
Progression from initial codes to themes.

Initial Codes	Intermediate Clustering	Final Themes
Internal funding for innovation Resource constraints on innovation Human capital limitations	Internal resource factors	Internal Innovation Capacities
External collaboration challenges Partnership benefits Knowledge sharing dynamics	External relationship factors	External Cooperation and Innovation
Trust-driven innovation Contractual protection limitations Relational governance preference	Governance mechanism factors	Governance Mechanisms and Innovation Dynamics

Table 3
Comprehensive case study overview.

Aspect	Case 1	Case 2	Case 3
Company Profile			
Industry Focus	Woven fabrics for household linen, security clothing, and workwear	Woven knit fabrics for swimwear, sportswear, women's fashion, and lingerie	Home linen and sportswear
Employee Count	87 employees	156 employees	94 employees
Years in Operation	18 years	22 years	15 years
Subcontracting Profile			
Primary Markets	European Union, North America	European Union, Australia	European Union, Canada
Relationship Duration	8–12 years with main contractors	5–15 years with main contractors	6–10 years with main contractors
Contractor Countries	France, Germany, Netherlands	Italy, France, Spain	Belgium, France, Germany
Innovation Profile			
Primary Innovation Type	Process Innovation	Product Innovation	Process Innovation
Innovation Frequency	Continuous improvement	Project-based	Periodic enhancement
R&D Investment	3–5 % of revenue	8–12 % of revenue	2–4 % of revenue
Governance Characteristics			
Relationship Orientation	Trust-dominant	Balanced hybrid	Relationship-constrained
Contract Formalization	Medium	Medium-High	Low-Medium
Communication Frequency	Daily/Weekly	Weekly/Bi-weekly	Monthly/As-needed

balance between contractual governance (formal agreements, legal protections) and relational governance (trust, communication, mutual commitment) and their differential impacts on innovation behaviors, contributing to governance theory understanding.

3.8.5. Cross-Case pattern analysis

Systematic cross-case comparison was conducted to identify patterns that transcended individual case characteristics while respecting case-specific contexts. This analysis involved developing case-ordered matrices, thematic networks, and pattern-matching procedures to identify consistencies and variations across cases.

Pattern identification focused on both convergent themes (consistent across cases) and divergent patterns (varying by case characteristics) to develop a nuanced understanding of contextual factors influencing governance-innovation relationships.

3.8.6. Validation and reliability procedures

3.8.6.1. Methodological triangulation. Methodological triangulation enhanced the credibility and validity of our findings by cross-referencing interview data with observational field notes and secondary documentation, including industry reports and company records. This triangulation process allowed us to confirm consistencies, identify discrepancies, and strengthen our interpretations through multiple data source perspectives.

Triangulation involved systematic comparison of findings across data sources, with particular attention to areas where different sources provided conflicting or contradictory information. Such discrepancies were explored through additional analysis and, where possible, follow-up discussions with participants.

3.8.6.2. Member checking procedures. Member checking was conducted through two mechanisms: transcript verification and preliminary findings review. Participants reviewed their interview transcripts for accuracy and provided clarifications where necessary. Additionally, preliminary findings were shared with key participants to verify interpretation accuracy and gather additional insights.

Participant feedback was incorporated into the analysis where appropriate, while maintaining analytical independence and theoretical coherence. Disagreements between participant interpretations and analytical conclusions were documented and discussed in the findings where relevant.

3.8.6.3. Inter-coder reliability. The coding process employed collaborative procedures where initial codes were assigned independently by two researchers, then reviewed and refined through team discussions to ensure inter-subjective agreement. Inter-coder reliability was calculated at multiple stages, with Cohen's kappa coefficients exceeding 0.80 for major theme assignments.

Discrepancies in coding were resolved through discussion and re-examination of the data, with disagreements serving as opportunities for deeper analysis and understanding rather than simply seeking consensus. The iterative nature of the process allowed for refinement of coding frameworks based on analytical learning.

3.8.6.4. Data saturation assessment. The analysis continued until data saturation was achieved, determined when subsequent interviews and observations yielded recurring patterns without introducing novel concepts or themes. Saturation was assessed both within individual themes and across the overall analytical framework.

Evidence of saturation included: consistent pattern emergence across new data, diminishing returns from additional coding, and theoretical coherence across all major themes. The decision to conclude data collection and analysis was made collectively by the research team based on these saturation indicators.

3.8.7. Analytical quality assurance

3.8.7.1. Audit trail maintenance. A comprehensive audit trail was maintained throughout the analytical process, documenting analytical decisions, code development evolution, theme refinement processes, and interpretation rationales. This trail enables external review of analytical procedures and supports analytical transparency.

The audit trail included: original coding schemes, code definition evolution, theme development documentation, analytical memos, and decision justification records. This documentation provides evidence of analytical rigor and enables potential replication studies.

3.8.7.2. Reflexivity and bias management. Reflexivity procedures were implemented to acknowledge and manage researcher bias throughout the analytical process. Regular reflexivity sessions were conducted to

Table 4
Internal innovation capacities analysis.

Capacity Dimension	Case 1	Case 2	Case 3	Cross-Case Patterns
Financial Resources				
Innovation Fund	Dedicated fund (€15,000/year)	Flexible budget (€45,000/year)	Limited allocation (€8000/year)	Dedicated funding correlates with innovation success
R&D Investment	3.2 % of revenue	9.8 % of revenue	2.1 % of revenue	Higher investment enables more ambitious innovation.
External Funding	Limited access	Grant funding secured	Occasional grants	External funding supplements internal limitations
Human Resources				
Technical Expertise	6 specialized technicians	12-person R&D team	3 technical staff	Team size limits innovation, scope, and complexity
Training Investment	€2500/year per employee	€4200/year per employee	€1800/year per employee	Training investment enhances innovation capacity
Management Commitment	High leadership support	Very high commitment	Moderate support	Leadership commitment drives innovation culture
Organizational Capabilities				
Knowledge Management	Informal documentation	Systematic knowledge systems	Basic documentation	Systematic approaches enhance learning
Process Standardization	Moderate standardization	High standardization	Low standardization	Standardization enables consistent innovation
Learning Culture	Strong improvement focus	Innovation-oriented culture	Traditional approach	Culture significantly influences innovation outcomes

Table 5
External cooperation analysis.

Cooperation Dimension	Case 1	Case 2	Case 3	Innovation Impact
R&D Partnerships				
Consultant Relationships	Equipment suppliers, Process experts	Design consultants, Technical specialists	Limited consultancy	Higher engagement enhances innovation
Research Collaborations	Industry associations	University partnerships	Minimal collaboration	Academic links provide cutting-edge knowledge
Technology Partnerships	Equipment manufacturers	Material suppliers, Tech providers	Basic supplier relations	Strategic partnerships enable breakthrough innovation
Knowledge Sharing				
Information Transparency	Moderate openness	High transparency	Limited sharing	Transparency facilitates collaborative innovation
Joint Problem-Solving	Regular collaboration	Systematic joint projects	Occasional cooperation	Joint approaches accelerate innovation
Best Practice Exchange	Informal sharing	Formal exchange programs	Limited exchange	Systematic exchange enhances capabilities
Contractor Relationships				
Innovation Support	Mixed contractor attitudes	Strong innovation encouragement	Innovation discouragement	Contractor support critical for innovation
Resource Provision	Limited contractor support	Substantial resource sharing	Minimal contractor contribution	Resource sharing enables ambitious innovation
Risk Sharing	Partial risk sharing	Collaborative risk management	Risk concentration on SME	Risk sharing enables innovation investment

Table 6
Governance mechanisms and innovation outcomes.

Governance Dimension	Case 1	Case 2	Case 3	Innovation Impact Analysis
Contractual Governance				
Contract Formalization	Medium formalization	High formalization	Low formalization	Moderate formalization provides flexibility with protection
Legal Protection	Basic IP protections	Comprehensive protections	Limited protections	Strong protections enable innovation investment
Performance Metrics	Quality and delivery focus	Innovation and quality metrics	Traditional cost metrics	Innovation-oriented metrics drive behavior
Dispute Resolution	Informal resolution preferred	Formal mechanisms available	Limited formal options	Clear mechanisms reduce innovation risk
Relational Governance				
Trust Level	High mutual trust	Very high trust	Moderate trust	Trust enables risk-taking and collaboration
Communication Quality	Regular, open communication	Systematic, transparent	Limited, formal only	Open communication facilitates innovation
Commitment Strength	Strong mutual commitment	Very strong commitment	Weak commitment	Commitment sustains long-term innovation
Flexibility	High relationship flexibility	Very high flexibility	Limited flexibility	Flexibility enables adaptation and innovation
Hybrid Governance				
Balance Achievement	Trust-dominant hybrid	Balanced hybrid approach	Governance gaps present	Balance optimizes innovation outcomes
Dynamic Adjustment	Moderate adaptability	High adaptability	Limited adaptability	Adaptability enables innovation evolution
Conflict Resolution	Relationship-based resolution	Multiple resolution options	Conflict avoidance	Effective resolution maintains innovation focus

Table 7
Comprehensive innovation behaviors and performance analysis.

Innovation Aspect	Case 1	Case 2	Case 3	Cross-Case Analysis
Innovation Type and Focus				
Primary Innovation	Process innovation (cutting & assembly)	Product innovation (new lines)	Process innovation (fabric treatment)	Innovation type aligns with capabilities and governance
Innovation Frequency	Continuous improvement	Project-based cycles	Periodic enhancement	Frequency reflects resource availability and support
Innovation Scope	Operational efficiency	Market expansion	Quality improvement	The scope is limited by internal and external constraints
Resource Utilization				
Internal Resources	Innovation fund, training	R&D team, systematic processes	Limited funding, basic capabilities	Internal resources determine innovation ambition
External Resources	Equipment suppliers, consultants	Multiple partnerships, grants	Occasional external funding	External access multiplies innovation potential
Resource Integration	Moderate integration	High integration	Limited integration	Integration effectiveness determines success
Performance Outcomes				
Quantitative Results	15 % waste reduction, 10 % time savings	20 % revenue increase, 30 % portfolio expansion	25 % durability improvement, 30 % complaint reduction	Results align with innovation type and investment
Qualitative Benefits	Employee engagement, process standardization	Market differentiation, brand strengthening	Customer satisfaction, reputation enhancement	Qualitative benefits support long-term competitiveness
Sustainability	Moderate sustainability	High sustainability	Limited sustainability	Sustainability requires ongoing resource commitment

Table 8
Cross-case pattern analysis.

Pattern Category	Consistent Findings	Variation Factors	Theoretical Implications
Resource-Innovation Relationship	Dedicated innovation resources correlate with success	Resource allocation strategies vary	RBV theory confirmed: resources enable competitive advantage
Governance-Innovation Nexus	Hybrid governance optimizes innovation outcomes	Balance points differ by context	Complementarity between governance mechanisms
Trust-Performance Link	High trust enables ambitious innovation	Trust development processes vary	Relational governance is critical for innovation
External Cooperation Impact	External partnerships multiply innovation potential	Partnership configurations differ	Open innovation principles validated
Innovation Type Alignment	Success requires capability-innovation alignment	Innovation choices reflect constraints	Strategic fit is essential for performance

Table 9
Theoretical contributions and framework implications.

Theoretical Area	Key Contributions	Framework Implications	Future Research Directions
Theory of Planned Behavior	Subjective norms and perceived control are critical for innovation behavior.	The extended TPB applies to organizational innovation contexts	Test TPB extensions across different cultural contexts
Resource-Based View	Resource orchestration is more important than resource possession	Dynamic capabilities complement static resources	Examine resource orchestration mechanisms
Governance Theory	Contractual and relational governance are complementary for innovation	Hybrid governance frameworks are needed	Develop contingency models for governance choice
Innovation Theory	Innovation-type governance alignment is critical for success	Innovation strategies must align with governance capabilities	Explore innovation-governance configurations
Subcontracting Literature	Trust enables innovation and risk-taking in asymmetric relationships	Relational governance is essential for SME innovation	Examine governance evolution over the relationship lifecycle

examine how researchers' backgrounds, theoretical orientations, and cultural perspectives might influence data interpretation.

Potential biases were addressed through multiple researcher involvement, diverse theoretical perspective consideration, and systematic challenging of preliminary interpretations. The research team's interdisciplinary backgrounds provided multiple analytical lenses while maintaining focus on the research questions.

3.8.7.3. External review processes. External review was sought from experienced qualitative researchers not directly involved in the study to provide an independent assessment of analytical procedures and interpretations. These reviews focused on methodological rigor, theoretical coherence, and empirical support for major conclusions.

External reviewer feedback was incorporated where appropriate while maintaining analytical autonomy and theoretical consistency. The external review process enhanced confidence in analytical quality and provided additional perspectives on interpretation validity.

3.8.8. Theoretical integration and framework development

The final analytical phase involved integrating thematic findings with existing theoretical frameworks while identifying areas where empirical findings extended or challenged theoretical expectations. This integration process was guided by abductive reasoning that moved between empirical patterns and theoretical explanation.

Theoretical integration resulted in the development of an enhanced conceptual framework that synthesized the Theory of Planned Behavior and the Resource-Based View perspectives while incorporating governance theory insights. This framework provides the foundation for both the empirical findings presentation and the theoretical contribution discussion.

By anchoring our analysis in established theoretical frameworks while remaining open to emergent insights, we achieved a balance between theoretical grounding and empirical discovery that contributes to both theory building and practical understanding in SME innovation governance.

3.9. Limitations and delimitations

Our study acknowledges several limitations while maintaining focus on its specific scope and objectives. The study focuses on three subcontracting SMEs within Morocco's textile sector and does not seek statistical generalization, instead offering rich, context-specific insights into innovation processes and governance dynamics. Limitations include the small sample size limiting generalizability beyond similar contexts, a single industry focus on textile and clothing potentially limiting relevance to other sectors, geographic constraints to Morocco potentially limiting applicability to other emerging economies, temporal limitation with data collection in a specific period, potentially missing evolutionary changes, and language barriers potentially affecting nuanced understanding despite multilingual capabilities.

The study was deliberately bounded by specific delimitations that define its scope and focus. These delimitations include focus on SMEs within specific size parameters (10–200 employees) to maintain definitional consistency, emphasis on international subcontracting relationships rather than domestic arrangements, concentration on innovation behaviors rather than innovation outcomes to understand processes rather than results, and exclusion of micro-enterprises and large corporations to maintain focus on the SME context and its unique characteristics.

3.10. Quality assurance

Quality assurance measures were implemented throughout the research process to ensure credibility, transferability, dependability, and confirmability of findings. Credibility was enhanced through member checking, where participants reviewed transcripts and preliminary findings, prolonged engagement involving extended field time for deeper understanding, and triangulation using multiple data sources and methods to confirm findings and provide comprehensive perspectives.

Transferability was addressed through thick description, providing detailed contextual information enabling readers to assess applicability to other contexts, purposive sampling involving strategic case selection for analytical generalization, and clear documentation maintaining transparent methodology for potential replication. Dependability was ensured through maintaining an audit trail with systematic documentation of research decisions, external review involving independent examination of the research process, and consistent procedures using standardized protocols across cases.

Confirmability was achieved through reflexivity involving researcher bias acknowledgment and management, data verification through cross-checking of findings across sources, and clear documentation separating data, findings, and interpretations to maintain analytical transparency. By anchoring our analysis in established theoretical frameworks, particularly those concerning relational and contractual governance, we contribute to theory-building in SME innovation research and lay the groundwork for further studies in varied contexts and industries.

4. Findings

This section presents the comprehensive findings from our qualitative analysis of innovation behaviors in three Moroccan textile SMEs engaged in international subcontracting relationships. The findings are organized thematically based on our conceptual framework and emergent patterns from the data analysis. Rather than merely describing individual cases, we present theoretically grounded insights that illuminate the complex dynamics between governance mechanisms and innovation behaviors in subcontracting contexts.

Our thematic analysis revealed three primary themes that capture the essential dynamics of SME innovation behaviors: Internal Innovation Capacities, External Cooperation and Innovation, and Governance Mechanisms and Innovation Dynamics. Each theme encompasses multiple dimensions and reflects the complex interplay between internal capabilities, external relationships, and governance structures that characterize successful innovation in resource-constrained environments.

4.1. Case study overview and participant characteristics

This comprehensive overview reveals significant diversity across the three cases in terms of innovation approaches, market orientation, and governance configurations. The variation in employee count, operational experience, and innovation investment patterns provides a rich foundation for understanding how different organizational characteristics influence innovation behaviors and governance preferences in subcontracting relationships.

4.2. Theme one: internal innovation capacities

Internal innovation capacities emerged as the foundational theme influencing SMEs' ability to engage in innovation activities within subcontracting relationships. Like the root system of a tree, these internal capacities determine how effectively organizations can absorb external resources and withstand environmental pressures while maintaining growth and adaptation. Our analysis identified multiple dimensions of internal capacity that collectively shape innovation potential and performance.

4.2.1. Financial resources and innovation investment

The analysis revealed significant variations in financial resource allocation for innovation across the three cases, with profound implications for innovation scope and sustainability. Case 2 demonstrated the most robust financial commitment to innovation, allocating nearly 10 % of revenue to R&D activities and maintaining a flexible innovation budget of €45,000 annually. This substantial investment enabled the company to pursue ambitious product innovation projects, including the development of new swimwear and sportswear lines that generated significant market success. As the Technical Director explained, "Our investment in R&D allows us to take calculated risks and explore new opportunities that our competitors cannot afford to pursue."

In contrast, Case 1 and Case 3 operated with more constrained financial resources, though they approached these limitations differently. Case 1 established a dedicated innovation fund of €15,000 annually, which, while modest, provided predictable resources for continuous process improvements. This strategic approach to resource allocation enabled systematic enhancement of cutting and assembly operations, resulting in measurable efficiency gains. Case 3 faced the most severe financial constraints, with limited innovation allocation of only €8000 per year. The R&D responsible noted: "We often have good ideas, but we lack the financial resources to implement them properly, which forces us to focus only on the most essential improvements."

The financial resource analysis demonstrates a clear relationship between innovation investment levels and innovation ambition. Companies with dedicated, substantial funding could pursue more complex, higher-risk innovations, while those with limited resources necessarily focus on incremental improvements with predictable returns. However, the presence of any dedicated innovation funding, regardless of amount, appeared to foster innovation culture and systematic improvement approaches.

4.2.2. Human capital and technical expertise

Human capital emerged as arguably the most critical component of internal innovation capacity, with technical expertise, training investment, and management commitment serving as key determinants of innovation potential. Case 2′s 12-person R&D team represented a significant organizational commitment to innovation capability, enabling the company to manage complex product development projects while maintaining operational excellence. The team's diverse expertise in textile technology, fashion design, and market research created synergistic capabilities that exceeded the sum of individual contributions.

Case 1 demonstrated how strategic human resource development could overcome size limitations through focused technical expertise. The company's six specialized technicians, despite representing a smaller team than Case 2, achieved significant process innovation through deep specialization in production optimization and quality improvement. The Production Director explained: "Our technicians may be fewer in number, but their specialized knowledge of our processes allows them to identify improvement opportunities that generalists might miss."

Case 3′s limited technical staff of three individuals highlighted the constraints that human resource limitations place on innovation scope and complexity. While these individuals possessed valuable technical knowledge, their limited capacity restricted the company's ability to pursue multiple innovation initiatives simultaneously or to engage in complex collaborative projects with external partners. This resource constraint forced the company to be highly selective in innovation projects, focusing only on those with immediate operational benefits.

Training investment patterns revealed additional insights into human capital development approaches. Case 2′s substantial training investment of €4200 per employee annually supported continuous capability enhancement and knowledge updating, enabling the organization to maintain technological currency and adapt to evolving market requirements. Case 1′s moderate investment of €2500 per employee supported targeted skill development aligned with process improvement objectives, while Case 3′s limited investment of €1800 per employee constrained capability development and innovation potential.

4.2.3. Organizational learning and knowledge management

Organizational learning capabilities varied significantly across the three cases, with systematic approaches to knowledge management correlating strongly with innovation success and sustainability. Case 2 implemented comprehensive knowledge management systems that captured, codified, and shared innovation-related learning across organizational levels. These systems included formal documentation of innovation processes, systematic capture of lessons learned, and structured knowledge sharing mechanisms that enabled cumulative learning and capability building.

Case 1 employed more informal but nonetheless effective knowledge management approaches, with strong emphasis on experiential learning and knowledge sharing through direct interaction and mentoring. While lacking formal documentation systems, the organization maintained institutional memory through experienced personnel and established communication networks that facilitated knowledge transfer and continuous improvement.

Case 3′s basic documentation and traditional knowledge management approaches limited organizational learning and innovation accumulation. The absence of systematic knowledge capture mechanisms meant that innovation insights and lessons learned were often lost or remained confined to individual experience rather than becoming organizational capabilities.

4.3. Theme two: external cooperation and innovation

External cooperation emerged as a critical mechanism through which SMEs transcended internal resource limitations and accessed capabilities necessary for innovation success. Like bridges connecting isolated islands, external cooperation relationships enabled SMEs to access knowledge, resources, and opportunities that would otherwise remain beyond their reach. However, these relationships also introduced complexity, risk, and dependency that required careful management to realize their innovation potential.

4.3.1. R&D partnerships and technical collaboration

External R&D partnerships demonstrated significant variation in scope, sophistication, and innovation impact across the three cases. Case 2 established the most comprehensive external collaboration network, including design consultants, technical specialists, university partnerships, and strategic technology providers. These relationships enabled the company to access cutting-edge knowledge and capabilities that would have been impossible to develop internally, given resource constraints. The Technical Director noted: "Our partnerships allow us to punch above our weight - we can access expertise and technologies that companies many times our size would struggle to maintain internally."

Case 2′s university partnerships proved particularly valuable for accessing research-based knowledge and emerging technologies. Through collaboration with textile engineering programs, the company gained access to the latest research findings, testing facilities, and graduate-level technical expertise. These relationships also provided opportunities for joint research projects that advanced both academic knowledge and practical innovation capabilities.

Case 1 maintained more focused but nonetheless valuable external partnerships, primarily with equipment suppliers and process optimization experts. While less extensive than Case 2′s network, these relationships provided targeted expertise for specific innovation challenges and enabled access to best practices from other industries and contexts. The owner explained: "We may not have the resources for extensive partnerships, but we've built strong relationships with key experts who understand our business and can provide practical solutions."

Case 3′s limited external collaboration reflected both resource constraints and organizational orientation toward self-reliance. The company's minimal engagement with consultants and research institutions limited access to external knowledge and capabilities, constraining innovation scope and sophistication. However, even limited external relationships proved valuable when properly leveraged, as demonstrated by the company's occasional grant-funded innovation projects.

4.3.2. Knowledge sharing and information exchange

Knowledge-sharing practices varied dramatically across cases and significantly influenced innovation outcomes and relationship sustainability. Case 2 demonstrated the most open and systematic approach to knowledge sharing, establishing formal exchange programs with contractors and partners while maintaining appropriate intellectual property protections. This transparency facilitated collaborative innovation and joint problem-solving while building trust and mutual commitment among relationship partners.

The company's approach to knowledge sharing extended beyond technical information to include market insights, trend analysis, and strategic planning discussions. This comprehensive information exchange enabled more effective innovation planning and reduced market risk through shared intelligence and collaborative market analysis. As one contractor representative noted during our observations: "Working with [Case 2] feels like a true partnership - they share information that helps us all succeed rather than hoarding knowledge for competitive advantage."

Case 1 employed more selective but nonetheless effective knowledge sharing approaches, focusing on operational information and process improvements while maintaining confidentiality around sensitive business information. This balanced approach enabled collaborative problem-solving and continuous improvement while protecting competitive advantages and maintaining appropriate boundaries in contractor relationships.

Case 3′s limited knowledge sharing reflected both resource constraints and trust limitations in contractor relationships. Restricted information exchange hindered collaborative innovation and problem-solving while reinforcing transactional rather than relational approaches to contractor partnerships. The limited transparency also constrained the company's ability to access contractor knowledge and capabilities that could have enhanced innovation potential.

4.3.3. Contractor support and resource provision

Contractor support for innovation activities varied significantly across cases and proved to be a critical determinant of innovation success and sustainability. Case 2 benefited from substantial contractor support, including financial contributions to innovation projects, resource sharing for research and development, and collaborative risk management approaches. This support enabled more ambitious innovation projects while reducing financial risk and resource constraints that typically limit SME innovation.

Contractors' willingness to share innovation risks and provide resources reflected the strength of relational governance and mutual trust developed over years of successful collaboration. One contractor representative explained: "We invest in [Case 2′s] innovation capabilities because their success directly contributes to our competitiveness and market success. It's a win-win relationship that benefits everyone involved."

Case 1 experienced mixed contractor attitudes toward innovation, with some contractors encouraging and supporting improvement initiatives while others emphasized cost control and traditional approaches. This variability required careful relationship management and selective innovation project planning to align with contractor preferences and expectations. The owner noted: "We've learned to read our contractors' attitudes toward innovation and adjust our approach accordingly - some want cutting-edge improvements while others prefer proven, cost-effective solutions."

Case 3 faced the most challenging contractor relationships regarding innovation support, with contractors generally discouraging innovation initiatives and emphasizing cost minimization over improvement and development. This constraint significantly limited the company's innovation ambitions and forced a focus on internal process improvements rather than collaborative innovation projects. The R&D responsible explained: "Sometimes we feel like contractors see innovation as unnecessary risk rather than opportunity - they want consistency and low cost above everything else."

4.4. Theme three: governance mechanisms and innovation dynamics

Governance mechanisms emerged as the orchestrating force that either harmonized or created discord among various elements of the innovation ecosystem. Like a conductor leading an orchestra, effective governance aligned different components - internal capabilities, external partnerships, resource allocation, and risk management - to create innovative outcomes. Our analysis revealed complex interactions between contractual and relational governance approaches, with different configurations yielding varying innovation results.

4.4.1. Contractual governance and innovation framework

Contractual governance provided essential structural foundations for innovation activities, though its effectiveness varied significantly based on formalization level, scope, and alignment with relational dynamics. Case 2 demonstrated the most sophisticated contractual governance approach, incorporating comprehensive intellectual property protections, innovation-oriented performance metrics, and clear dispute resolution mechanisms. These contractual elements created a framework that encouraged innovation investment while protecting all parties' interests and contributions.

The company's contracts explicitly included innovation-related clauses, intellectual property sharing agreements, and collaborative development provisions that formalized innovation partnerships while maintaining appropriate protections. This approach enabled contractors and subcontractors to invest in joint innovation projects with confidence in their rights and returns. The General Manager explained: "Our contracts provide the foundation for innovation partnerships - they define roles, responsibilities, and rewards clearly so everyone can focus on creating value rather than protecting their interests."

Case 1 employed moderate contractual formalization that balanced protection with flexibility, enabling innovation activities within established relationship frameworks while maintaining operational adaptability. The company's contracts focused primarily on quality and delivery requirements while allowing informal innovation, collaboration, and improvement initiatives. This approach provided sufficient structure for operational excellence while preserving relational flexibility for innovation and adaptation.

Case 3′s limited contractual governance created vulnerability and uncertainty that constrained innovation investment and collaborative development. The absence of clear intellectual property protections and innovation-related provisions discouraged both internal innovation investment and external collaboration, limiting the company's ability to pursue ambitious improvement initiatives or develop strategic partnerships.

4.4.2. Relational governance and trust dynamics

Relational governance emerged as particularly crucial for innovation success, with trust, communication quality, and mutual commitment serving as fundamental enablers of collaborative innovation and risk-taking. Case 2 demonstrated exceptional relational governance characterized by very high trust levels, systematic, transparent communication, and strong mutual commitment between contractors and subcontractors. These relational dynamics enabled ambitious innovation projects, resource sharing, and collaborative risk management that would have been impossible under purely contractual arrangements.

The development of trust in Case 2 occurred gradually through consistent performance, open communication, and mutual support during challenging periods. This trust foundation enabled contractors to share sensitive market information, provide resources for innovation projects, and collaborate on strategic planning initiatives. The Technical Director noted: "Trust changes everything - when contractors trust us, they share information and resources that help us innovate better, and when we trust them, we're willing to invest in improvements that benefit the whole relationship."

Trust also enabled effective risk-taking and experimentation in innovation projects. Contractors' confidence in Case 2′s capabilities and commitment allowed for innovation projects with uncertain outcomes and longer payback periods, while the SME's trust in contractor support encouraged investment in capability development and process improvement beyond immediate operational requirements.

Case 1 maintained strong relational governance characterized by high mutual trust and regular, open communication, though with somewhat less systematic structure than Case 2. These relationships enabled continuous process improvement and collaborative problem-solving while maintaining operational flexibility and responsiveness. The informal nature of many relationships actually enhanced innovation by reducing bureaucratic constraints and enabling rapid adaptation and improvement.

Case 3′s moderate trust levels and limited communication quality constrained relational governance and innovation potential. Weak mutual commitment and limited relationship flexibility restricted collaborative innovation and discouraged investment in improvement initiatives beyond immediate operational requirements. The constrained relational dynamics reinforced transactional approaches rather than fostering the collaborative partnerships necessary for sustained innovation.

4.4.3. Hybrid governance and dynamic balance

The most successful innovation outcomes occurred when SMEs achieved an effective balance between contractual structure and relational flexibility, creating hybrid governance approaches that provided protection while enabling collaborative dynamics necessary for innovation. Case 2 exemplified this balanced hybrid approach, combining comprehensive contractual frameworks with strong relational governance to optimize innovation potential while managing risks and protecting interests.

This hybrid governance enabled dynamic adjustment and adaptation as innovation projects evolved and relationship requirements changed. The combination of clear contractual foundations and strong relational trust allowed for flexibility in project scope, timeline adjustments, and resource allocation modifications without undermining relationship stability or partner confidence.

Case 1 achieved a trust-dominant hybrid approach that prioritized relational governance while maintaining sufficient contractual structure for protection and clarity. This approach proved effective for continuous improvement and process innovation, though it may have limited potential for more complex or ambitious innovation projects requiring substantial resource investment and risk sharing.

Case 3′s governance gaps and limited integration between contractual and relational elements constrained innovation potential and relationship development. The absence of effective hybrid governance limited the company's ability to balance protection with collaboration, constraining both innovation ambition and partnership development.

4.5. Innovation behaviors and outcomes

4.5.1. Innovation type differentiation and strategic alignment

The analysis revealed clear differentiation between process and product innovation approaches, with each requiring distinct resource configurations, governance mechanisms, and risk management strategies. Process innovation, exemplified by Cases 1 and 3, focused on operational efficiency improvements, quality enhancement, and cost reduction through internal capability development and targeted external collaboration. These innovations typically required moderate resource investment, generated predictable short-term returns, and aligned well with incremental improvement cultures and resource-constrained environments.

Product innovation, demonstrated by Case 2, required substantially higher resource investment, more sophisticated external partnerships, and greater risk tolerance, but offered potential for significant competitive advantage and market expansion. The company's development of new swimwear and sportswear lines required comprehensive market research, design expertise, technical development capabilities, and marketing support that exceeded typical SME internal resources. However, successful product innovation generated substantial revenue growth and market differentiation that justified the investment and risk.

The alignment between innovation type and organizational capabilities proved critical for success. Cases that matched innovation approaches to their resource capabilities, governance strengths, and market opportunities achieved better outcomes than those attempting innovations beyond their capabilities or support structures. This alignment consideration has important implications for SME innovation strategy and resource allocation decisions.

4.5.2. Performance measurement and outcome assessment

Performance outcomes varied significantly across cases, reflecting differences in innovation type, resource investment, and governance effectiveness [111]. Case 1′s process innovation achievements included measurable operational improvements: 15 % reduction in material waste, 10 % decrease in assembly time, and 8 % improvement in overall productivity. These results generated immediate cost savings and efficiency gains while building organizational capabilities for future improvement initiatives.

Beyond quantitative improvements, Case 1 achieved significant qualitative benefits, including enhanced employee engagement through involvement in improvement initiatives, improved process standardization that reduced variation and errors, and strengthened contractor relationships through demonstrated commitment to operational excellence. These qualitative benefits created foundations for sustained competitive advantage and continued innovation.

Case 2′s product innovation generated the most substantial performance improvements: 20 % increase in revenue attributed to new product lines, 30 % expansion in product portfolio breadth, and 12 % improvement in profit margins through higher value-added offerings. These quantitative results demonstrated the potential returns from successful product innovation while validating the company's substantial investment in R&D capabilities and external partnerships.

The qualitative benefits for Case 2 included significant market differentiation through unique product offerings, enhanced brand positioning in target markets, and stronger customer loyalty through innovative product features. These benefits created competitive advantages that extended beyond immediate financial returns to establish market position and relationship strength for future growth.

Case 3′s more limited innovation achievements included 25 % improvement in fabric durability through advanced treatment techniques, 30 % reduction in customer complaints related to quality issues, and 18 % increase in customer satisfaction scores. While more modest than other cases, these improvements demonstrated the value of innovation even within constrained resource environments.

4.6. Cross-case analysis and pattern identification

4.6.1. Resource configuration and innovation performance

Cross-case analysis revealed consistent patterns linking resource configuration to innovation performance, while highlighting important variations in resource allocation strategies and optimization approaches. All three cases demonstrated that dedicated innovation resources, regardless of absolute amount, generated better innovation outcomes than ad hoc or reactive approaches to innovation investment. However, the optimal resource configuration varied based on innovation type, organizational capabilities, and external support availability.

Case 2′s substantial resource investment in human capital, particularly the 12-person R&D team, enabled ambitious product innovation that generated substantial returns. However, this approach required significant ongoing investment and sophisticated management capabilities that may not be feasible for all SMEs. Case 1′s more modest but systematic resource allocation achieved significant process improvements through focused investment in training and equipment, demonstrating that strategic resource application could generate innovation success even with limited resources.

The analysis also revealed the multiplicative effect of combining internal and external resources through partnerships and collaboration. Cases that effectively integrated internal capabilities with external expertise and resources achieved innovation outcomes that exceeded what either internal or external resources could have generated independently. This finding supports open innovation principles and highlights the importance of resource orchestration capabilities for SME innovation success.

4.6.2. Governance mechanism complementarity

Cross-case analysis strongly supported the complementarity rather than substitutability of contractual and relational governance mechanisms in fostering innovation. All cases that achieved strong innovation outcomes employed hybrid governance approaches that combined appropriate contractual frameworks with strong relational dynamics. However, the optimal balance point between contractual and relational elements varied based on innovation type, relationship maturity, and environmental uncertainty.

Product innovation appeared to require more sophisticated

contractual governance due to higher intellectual property risks and more complex resource sharing requirements, while process innovation could succeed with simpler contractual frameworks supported by strong relational governance. However, both innovation types benefited from trust-based relationships that enabled risk-taking, knowledge sharing, and collaborative problem-solving.

The dynamic nature of governance balance also emerged as important, with successful cases demonstrating the ability to adjust governance approaches as relationships evolved and innovation requirements changed. This adaptability enabled sustained innovation over time while maintaining relationship stability and partner satisfaction.

4.7. Theoretical implications and framework refinement

4.7.1. Extended theory of planned behavior in organizational innovation

Our findings significantly extend the Theory of Planned Behavior from individual to organizational contexts, demonstrating how subjective norms (contractor expectations), attitudes (innovation orientation), and perceived behavioral control (resource availability and governance support) influence SME innovation behaviors. The analysis revealed that contractor attitudes and expectations regarding innovation create powerful subjective norms that either encourage or discourage SME innovation investment, with Cases 1 and 2 benefiting from contractor support while Case 3 faced innovation discouragement.

Perceived behavioral control emerged as particularly important in resource-constrained SME contexts, with internal capabilities and external support significantly influencing innovation confidence and investment decisions. The interaction between internal resources and external governance in shaping perceived control provides important insights for understanding SME innovation behavior and designing support interventions.

4.7.2. Dynamic resource-based view and innovation capabilities

The findings enhance Resource-Based View theory by demonstrating that resource orchestration capabilities - the ability to combine, coordinate, and leverage internal and external resources - may be more important than absolute resource levels for SME innovation success. Case 2′s superior resource orchestration through systematic partnership management and capability integration enabled innovation outcomes that exceeded what its resources would suggest, while Case 3′s limited orchestration capabilities constrained innovation despite reasonable resource availability.

This insight suggests that SME innovation support should focus not only on providing resources but also on developing resource orchestration capabilities through training, networking, and governance support. The dynamic nature of resource requirements across innovation stages also emerged as important, with successful cases demonstrating the ability to reconfigure resources as innovation projects evolved.

4.7.3. Governance theory and innovation enablement

The analysis makes significant contributions to governance theory by demonstrating the complementary nature of contractual and relational governance for innovation in asymmetric relationships. Rather than substitutes, our findings show that contractual and relational governance mechanisms work together to create innovation-enabling environments that balance protection with collaboration, structure with flexibility, and risk management with opportunity pursuit.

The contingent nature of optimal governance configurations based on innovation type, relationship characteristics, and environmental factors provides important theoretical insights for governance design and management. The finding that governance balance points evolve dynamically as relationships and innovation requirements change adds temporal dimensions to governance theory that have been underexplored in previous research.

This comprehensive findings section provides a thorough analysis of the complex dynamics governing SME innovation behaviors in subcontracting relationships. The integration of multiple data sources, systematic thematic analysis, and cross-case comparison reveals important patterns and relationships that advance both theoretical understanding and practical knowledge in this critical area of organizational innovation and governance.

5. Discussion

This section critically examines our findings in relation to existing literature, highlighting convergences, divergences, and novel contributions to understanding SME innovation behaviors in subcontracting relationships. We position our results within the broader theoretical landscape while drawing precise implications for theory, practice, and policy. Think of this discussion as a bridge connecting our specific findings to the wider world of knowledge - it shows how our discoveries either support, challenge, or extend what we already know about innovation and governance in SME contexts.

5.1. Comparison and contrast with current literature

Our findings fundamentally challenge the prevailing view in governance literature that treats contractual and relational governance as substitutable mechanisms. While seminal works by Um and Oh [78] and Howard et al. [77] have predominantly conceptualized these governance forms as alternatives where firms choose one over the other based on contextual factors, our evidence demonstrates their essential complementarity for fostering innovation in SME subcontracting relationships. This represents a significant theoretical advancement that aligns with emerging perspectives from [120] and Cao and Lumineau [114], who advocate for hybrid governance approaches in complex supply networks. Our Case 2 exemplifies this complementarity, where comprehensive contractual frameworks provided the structural foundation for innovation investments while strong relational governance enabled the collaborative dynamics necessary for successful product development. This finding directly contradicts traditional transaction cost economics predictions that high trust should reduce the need for formal contracts, instead demonstrating that in innovation contexts characterized by high uncertainty and knowledge intensity, both governance mechanisms work synergistically to create innovation-enabling environments that balance protection with collaboration.

This complementarity perspective extends recent work by Wu et al. [18] and Song et al. [17], who examined trust-dependence relationships in construction contexts, but our study uniquely demonstrates how this complementarity operates specifically in innovation contexts within emerging economy SMEs. We reveal that the optimal balance between contractual and relational elements varies dynamically based on innovation type, relationship maturity, and resource availability, with product innovation appearing to require more sophisticated contractual governance due to higher intellectual property risks, while process innovation can succeed with simpler contractual frameworks supported by strong relational governance. This dynamic nature of governance balance represents an important advancement over static conceptualizations that dominate existing literature.

Our findings significantly advance Resource-Based View theory by demonstrating that resource orchestration capabilities may be more critical than absolute resource levels for SME innovation success. This insight challenges traditional RBV applications that focus primarily on resource possession [112] and aligns with contemporary dynamic capabilities perspectives. Case 2′s superior innovation outcomes, despite operating in the same resource-constrained environment as other cases, illustrate the power of effective resource orchestration, where the company's ability to systematically combine internal R&D capabilities with external partnerships, university collaborations, and contractor resources created innovation potential that exceeded what any individual resource component could have generated independently. This

finding resonates with Parida et al. [119] work on absorptive capacity but extends it by demonstrating how governance mechanisms mediate the relationship between resource orchestration and innovation outcomes. Conversely, Case 3′s limited innovation achievements despite reasonable resource availability highlight how poor orchestration capabilities can constrain innovation potential.

The critical role of trust in enabling innovation risk-taking emerges strongly from our findings, providing robust empirical support that extends social capital theory into SME subcontracting contexts. While Colquitt et al. [65] established the general relationship between trust and risk-taking behavior, our study demonstrates the specific mechanisms through which trust operates in asymmetric power relationships characteristic of subcontracting arrangements. The trust-innovation relationship proves particularly pronounced in our findings, with Case 2′s very high trust levels enabling ambitious product innovation projects with uncertain outcomes and longer payback periods, supporting recent work by Schoenherr et al. [12] on trust formation in outsourcing relationships while extending it by showing how trust specifically enables innovation investment decisions that rational economic actors might otherwise avoid due to risk and uncertainty. Our findings also reveal the temporal dimension of trust development, showing how consistent performance, open communication, and mutual support during challenging periods gradually build the trust foundation necessary for collaborative innovation, contributing to recent work by Heenkenda et al. [55] and Jabeen et al. [67] on innovation capabilities in SMEs while uniquely demonstrating how trust development interacts with governance mechanisms to create innovation-enabling environments.

Our application and extension of the Theory of Planned Behavior to organizational innovation contexts represents a significant theoretical contribution that addresses calls for broader TPB applications beyond individual behavior. While TPB has been primarily applied to individual technology adoption and entrepreneurial intentions [43,44], our study demonstrates its explanatory power for organizational innovation behaviors in subcontracting relationships. Our findings reveal how subjective norms (contractor expectations regarding innovation), attitudes (organizational innovation orientation), and perceived behavioral control (resource availability and governance support) collectively influence SME innovation behaviors, with Case 2′s strong innovation performance correlating with positive contractor attitudes toward innovation, strong organizational innovation orientation, and high perceived behavioral control through robust governance support. This extends recent work by Li et al. [36] on TPB applications in construction contexts while uniquely demonstrating the mechanisms through which these psychological factors operate in innovation decisions. The perceived behavioral control dimension emerges as particularly important in our resource-constrained SME context, where internal capabilities and external governance support significantly influence innovation confidence and investment decisions, contributing to recent work by Fang et al. [35] while extending it by showing how governance mechanisms specifically shape perceived control over innovation outcomes in subcontracting relationships.

Our clear differentiation between process and product innovation approaches and their distinct governance requirements contributes to ongoing debates in innovation management literature about innovation typologies and strategic alignment. While traditional distinctions between incremental and radical innovation focus primarily on technological characteristics, our study demonstrates how innovation type alignment with organizational capabilities and governance structures determines success. Our findings support recent work by Kahn [105] [105] on innovation understanding but extend it by demonstrating the specific governance configurations required for different innovation types in SME subcontracting contexts, with process innovation thriving under trust-dominant governance with moderate contractual formalization, while product innovation requires more balanced hybrid governance with sophisticated contractual protections and strong relational dynamics. This alignment perspective also contributes to recent work by Teixeira et al. [106] on innovation in family businesses and extends capability-based perspectives on innovation strategy, suggesting that SMEs should align innovation strategies not only with internal capabilities but also with governance strengths and external relationship characteristics, representing a multi-dimensional alignment perspective that has been underexplored in existing literature.

5.2. Theoretical and practical implications

Our findings contribute a dynamic complementarity framework to governance theory that transcends the traditional substitution versus complementarity debate, positing that optimal governance configurations for innovation involve dynamic balance points between contractual and relational mechanisms that evolve based on innovation type, relationship maturity, and environmental uncertainty. This framework addresses theoretical gaps identified by recent meta-analyses and provides a more nuanced understanding of governance choice in innovation contexts, suggesting that governance mechanisms operate as dynamic complements rather than static substitutes, with their effectiveness depending on temporal factors, innovation requirements, and relationship characteristics. This temporal dimension adds important theoretical depth to governance literature and provides a foundation for future research on governance evolution in innovation partnerships.

The extension of Resource-Based View theory emerges through our positioning of resource orchestration capabilities as potentially more important than resource possession for innovation success in resource-constrained environments. This extension challenges traditional RBV applications and contributes to dynamic capabilities literature by demonstrating the specific mechanisms through which orchestration operates in SME innovation contexts. The orchestration capabilities framework that emerges from our findings includes partnership management, capability integration, and governance configuration abilities that enable SMEs to leverage external resources for innovation purposes, providing a foundation for future research on resource strategy in innovation contexts while offering practical guidance for SME capability development. In addition, our findings contribute a governance-innovation configuration model that demonstrates how different innovation types require distinct governance approaches for success, extending innovation management literature by explicitly incorporating governance considerations into innovation strategy frameworks and providing a foundation for contingency approaches to innovation management in SME contexts.

For SME managers navigating complex governance decisions in subcontracting relationships, our findings provide actionable strategic guidance demonstrating that successful innovation requires systematic investment in both internal capabilities and external relationship management, with particular attention to governance configuration alignment with innovation objectives. SME managers should prioritize dedicated innovation funding and systematic capability development over ad hoc innovation approaches, as our evidence demonstrates that even modest but systematic resource allocation generates better outcomes than larger but inconsistent investments, suggesting that sustainability and predictability of innovation investment may be more important than absolute funding levels for resource-constrained SMEs. Managers should develop hybrid governance approaches that combine appropriate contractual protections with strong relational dynamics rather than relying exclusively on either formal contracts or informal relationships, with the specific balance point aligning with innovation type - product innovation requiring more sophisticated contractual frameworks and process innovation thriving under trust-dominant approaches with moderate formalization.

The systematic development of external partnership portfolios that provide access to complementary capabilities while maintaining appropriate intellectual property protections emerges as critical for SME success. Case 2′s success with university partnerships, design consultants, and technology providers demonstrates the multiplicative effect of

well-orchestrated external relationships on innovation potential, suggesting that SME managers should systematically develop these partnerships rather than pursue ad hoc external relationships. For managing contractor relationships to optimize innovation outcomes, the evidence demonstrates that contractor support and attitudes toward innovation significantly influence SME innovation behaviors and outcomes, suggesting that relationship management should explicitly incorporate innovation considerations. SME managers should proactively communicate innovation capabilities and interests to contractors while demonstrating how innovation contributes to mutual value creation, negotiate innovation risk-sharing arrangements with contractors rather than absorbing all innovation risks internally, and work with contractors to develop performance metrics that explicitly recognize and reward innovation contributions rather than focusing exclusively on cost and delivery metrics.

Our findings provide a framework for developing innovation strategies that align with SME capabilities and governance strengths while leveraging external relationships for enhanced innovation potential. SMEs should select innovation approaches that align with internal capabilities, governance strengths, and external support availability rather than pursuing innovation types beyond their orchestration capabilities, with process innovation potentially more appropriate for SMEs with limited resources but strong operational capabilities, while product innovation requires substantial resource investment and sophisticated partnership management. Managers should prioritize human capital development and knowledge management capabilities as foundations for innovation success, as our findings demonstrate that technical expertise, training investment, and systematic learning approaches significantly influence innovation potential regardless of absolute resource levels. SMEs should also develop systematic approaches to external knowledge integration that balance openness with intellectual property protection, with Case 2′s formal exchange programs and structured collaboration approaches demonstrating how systematic external integration can enhance innovation outcomes while managing associated risks.

For policymakers designing SME innovation support programs, our findings suggest that effective support should focus on capability building and ecosystem development rather than simply providing financial resources. Policy programs should include training components that help SME managers develop resource orchestration capabilities, including partnership management, capability integration, and governance configuration skills, as our findings demonstrate that these orchestration capabilities may be more important than absolute resource levels for innovation success. Policymakers should develop model governance frameworks and legal templates that help SMEs establish appropriate contractual foundations for innovation partnerships while maintaining relational flexibility, reducing transaction costs and risks associated with innovation collaboration, and enabling more SMEs to pursue ambitious innovation projects. Policy interventions should focus on creating innovation ecosystems that connect SMEs with universities, research institutions, and technical consultants rather than simply providing direct funding, as our findings demonstrate the multiplicative effect of external partnerships on innovation potential and suggest that ecosystem development may be more effective than individual firm support.

Industry associations should develop programs that encourage relational governance practices and collaborative innovation within subcontracting networks, including best practice sharing, networking events, and training programs that help firms develop trust-based relationships and collaborative problem-solving capabilities. Industry bodies should develop governance standards and best practices for innovation partnerships that help firms establish an appropriate balance between contractual protection and relational flexibility, reducing uncertainty and transaction costs while encouraging innovation investment. Industry programs should explicitly recognize and reward innovation contributions within subcontracting networks rather than focusing exclusively on cost and efficiency metrics, creating positive incentives for innovation investment and helping shift industry culture toward innovation-oriented partnerships.

Regulatory frameworks should provide accessible and affordable intellectual property protections that enable SMEs to pursue collaborative innovation while protecting their contributions, including simplified patent processes, collaborative IP frameworks, and dispute resolution mechanisms tailored to SME needs. Legal systems should provide efficient and affordable contract enforcement mechanisms that give SMEs confidence in formal governance arrangements while supporting relational approaches to dispute resolution, encouraging both contractual formalization and relational trust development. Tax and regulatory incentives should explicitly support collaborative innovation and knowledge sharing within subcontracting networks rather than focusing exclusively on individual firm innovation, encouraging the development of innovation-oriented partnerships and ecosystem approaches to innovation development.

Our discussion reveals several convergent themes that highlight the integrated nature of governance, resources, and innovation in SME subcontracting contexts, demonstrating that successful innovation requires alignment across multiple dimensions, including internal capabilities, external relationships, governance configurations, and environmental support. The dynamic nature of these relationships emerges as a critical insight, with successful SMEs demonstrating the ability to adapt governance approaches, reconfigure resources, and evolve partnerships as innovation requirements change, representing a meta-capability that enables sustained innovation over time and constitutes an important area for future research and practice development. The contextual sensitivity of our findings also highlights the importance of understanding innovation and governance within specific economic, cultural, and institutional environments, where, while our Moroccan textile industry context provides valuable insights, the transferability of specific findings to other contexts requires careful consideration of environmental factors and institutional differences. This comprehensive discussion positions our findings within the broader literature while highlighting their novel contributions to understanding SME innovation behaviors in subcontracting relationships, with the precise implications providing actionable guidance for managers, policymakers, and researchers while establishing foundations for future inquiry in this critical area of organizational innovation and governance.

6. Conclusion

This exploratory study has illuminated the complex dynamics governing innovation behaviors in SME subcontracting relationships, providing crucial insights into how governance mechanisms shape innovation outcomes in resource-constrained environments. Through our examination of three Moroccan textile SMEs engaged in international subcontracting, we have uncovered a nuanced ecosystem where innovation emerges not from isolated organizational capabilities, but from the intricate interplay between internal capacities, external cooperation, and governance configurations. Like a complex symphony where individual instruments must harmonize to create beautiful music, our findings reveal that successful SME innovation requires careful orchestration of multiple elements working in dynamic equilibrium.

Our central contribution challenges the prevailing theoretical assumption that contractual and relational governance mechanisms operate as substitutes in inter-organizational relationships. Instead, we demonstrate their essential complementarity for fostering innovation, particularly in contexts characterized by power asymmetries and resource constraints typical of SME subcontracting arrangements. This complementarity operates dynamically, with optimal balance points evolving based on innovation type, relationship maturity, and environmental conditions. Product innovation demands sophisticated contractual frameworks complemented by strong relational trust, while process innovation thrives under trust-dominant governance with

moderate contractual formalization. This finding fundamentally advances governance theory by introducing temporal and contextual dimensions that have been largely overlooked in existing literature.

The study reveals that resource orchestration capabilities, the ability to combine, coordinate, and leverage internal and external resources, may be more critical for SME innovation success than absolute resource levels. This insight significantly extends Resource-Based View theory beyond its traditional focus on resource possession toward a more dynamic understanding of how SMEs can transcend resource limitations through strategic governance and partnership management. Our evidence demonstrates that SMEs with superior orchestration capabilities achieve innovation outcomes that exceed what their individual resources would predict, while those with poor orchestration remain constrained despite reasonable resource availability.

Trust emerges as the cornerstone of innovation-enabling relationships, serving as both a facilitator of risk-taking behavior and a mechanism for knowledge sharing in uncertain environments. However, our findings reveal trust as a fragile and context-dependent resource that requires continuous cultivation through consistent performance, transparent communication, and mutual support during challenging periods. When trust erodes, as observed in Case 3, innovation ambitions become constrained regardless of internal capabilities or resource availability. This highlights the critical importance of relationship management in SME innovation strategies and suggests that trust-building should be viewed as a strategic investment rather than an operational byproduct.

The practical implications of our findings extend beyond academic understanding to provide actionable guidance for multiple stakeholder groups. SME managers should prioritize the development of hybrid governance approaches that balance contractual protection with relational flexibility, invest systematically in resource orchestration capabilities, and align innovation strategies with their governance strengths and external relationship characteristics. Contractors and supply chain partners should recognize subcontractors as potential innovation collaborators rather than mere service providers, developing risk-sharing arrangements and performance metrics that encourage rather than constrain innovation investment. Policymakers should focus on capability building and ecosystem development rather than simply providing financial resources, creating frameworks that facilitate collaborative innovation while protecting SME interests in asymmetric relationships.

6.1. Limitations and methodological considerations

While our study provides valuable insights into SME innovation behaviors in subcontracting contexts, several limitations must be acknowledged to properly contextualize our findings and guide future research directions. The qualitative design, while enabling deep exploration of complex phenomena, inherently limits the statistical generalizability of our results. Our focus on three SMEs, though allowing for rich contextual understanding and detailed analysis, restricts our ability to make broad claims about SME innovation behaviors across different organizational sizes, structures, or strategic orientations. The exploratory nature of our research was appropriate for initial theory building in this underexplored area, but confirmation of our propositions requires larger-scale empirical validation.

The geographic and sectoral specificity of our study presents both strengths and limitations for broader applicability. Our focus on Morocco's textile industry provides valuable insights into innovation dynamics in emerging economy contexts and traditional manufacturing sectors undergoing digital transformation pressures. However, this specificity raises questions about transferability to other cultural contexts, economic environments, institutional frameworks, and industrial sectors. The textile industry's particular characteristics, including relatively mature technology, established supply chains, and specific market dynamics, may influence governance-innovation relationships in ways that differ from high-technology industries, service sectors, or knowledge-intensive activities.

Temporal limitations also constrain our findings, as our data collection occurred during a specific period (September-October 2023) that may not capture the full evolution of governance-innovation relationships over longer time horizons. Innovation and governance are inherently dynamic processes that unfold over months and years, yet our cross-sectional approach provides only a snapshot of these relationships at a particular moment. This temporal constraint is particularly important given our emphasis on trust development and relationship evolution as critical factors in innovation enablement.

The language and cultural dimensions of our research, while carefully managed through multilingual capabilities and cultural sensitivity, may have influenced participant responses and our interpretation of findings. Despite conducting interviews in participants' preferred languages and employing cultural intermediaries, subtle nuances in meaning and context may have been lost in translation or cultural interpretation, potentially affecting the depth and accuracy of our understanding.

6.2. Future research directions

Our findings open several promising avenues for future research that could significantly advance understanding of governance-innovation relationships in SME contexts and beyond. These research directions range from validation studies to theoretical extensions and practical interventions.

6.2.1. Quantitative validation and scale studies

Large-scale quantitative studies examining governance mechanism complementarity across diverse industries, cultural contexts, and economic environments represent a critical next step for validating and extending our theoretical propositions. Such studies should employ sophisticated measurement approaches that capture the dynamic nature of governance configurations and their evolution over time, moving beyond static conceptualizations that dominate existing literature.

Survey-based research could test our complementarity hypothesis across hundreds of SME subcontracting relationships, examining whether the patterns we observed in Moroccan textiles hold across different sectors such as automotive, electronics, food processing, and knowledge-intensive services. Statistical modeling could quantify the relative contributions of contractual and relational governance to innovation outcomes while identifying the optimal balance points under different conditions.

Cross-cultural validation studies are particularly needed to understand how national business systems, cultural values, and institutional frameworks influence governance-innovation dynamics. Research comparing governance effectiveness across individualistic versus collectivistic cultures, high-trust versus low-trust societies, and formal versus informal institutional environments could illuminate the boundary conditions of our findings and contribute to a more nuanced theoretical understanding.

6.2.2. Longitudinal process studies

Longitudinal research examining the temporal evolution of governance-innovation relationships would provide crucial insights into how optimal governance configurations change as relationships mature, innovation requirements evolve, and environmental conditions shift. Such studies could illuminate the mechanisms through which trust develops in innovation partnerships, how governance rebalancing occurs in response to changing circumstances, and the long-term sustainability of different governance approaches under varying conditions.

Multi-year case studies tracking SME-contractor relationships from inception through maturity could reveal critical transition points where governance adjustments become necessary. These studies could examine how initial contractual arrangements evolve to accommodate increasing trust and collaborative capability, or conversely, how relationship deterioration affects governance mechanisms and innovation potential.

Process studies could also examine the innovation lifecycle within subcontracting relationships, investigating how governance needs differ across innovation stages from idea generation through implementation and commercialization. Understanding these temporal dynamics would provide valuable guidance for relationship management and governance design.

6.2.3. Sector-Specific and technology-mediated research

Cross-industry comparative studies could reveal how sector-specific characteristics influence governance-innovation dynamics and the transferability of governance configurations across different industrial contexts. Research comparing traditional manufacturing industries like textiles with high-technology sectors, knowledge-intensive services, and emerging digital industries could illuminate the contingency factors that determine optimal governance approaches.

The role of digital technologies in mediating governance mechanisms and enabling new forms of collaborative innovation represents another important research frontier. As digital platforms, artificial intelligence, and blockchain technologies reshape inter-organizational relationships, understanding how these technological developments interact with traditional governance mechanisms to influence innovation outcomes becomes increasingly critical for both theory and practice.

Studies examining virtual collaboration, digital trust-building, and technology-mediated knowledge sharing could reveal how governance mechanisms adapt to digital environments. Research on smart contracts, digital reputation systems, and automated governance mechanisms could contribute to understanding next-generation approaches to innovation collaboration.

6.2.4. Innovation ecosystem perspectives

While our study provides valuable insights into bilateral subcontracting relationships, future research should explore how these findings extend to broader innovation ecosystem contexts. Our analysis revealed complex interdependencies among internal capabilities, external relationships, and environmental factors, but the governance mechanisms operating in multi-actor innovation ecosystems may differ significantly from the bilateral relationships we examined.

Future research could investigate how SMEs navigate governance challenges when participating in innovation ecosystems involving multiple contractors, suppliers, research institutions, and support organizations simultaneously. The governance mechanisms that prove effective in dyadic subcontracting relationships may require adaptation or supplementation when applied to ecosystem contexts where multiple stakeholders have varying interests, power positions, and innovation objectives.

Ecosystem-level research could examine whether the complementarity between contractual and relational governance that we observed in bilateral relationships holds when extended to network contexts, or whether ecosystem participation requires entirely different governance approaches. Such research would contribute to understanding how policy interventions and industry initiatives could enhance SME innovation potential through ecosystem development while maintaining the trust-based relationships we found essential for innovation success.

Cross-ecosystem comparative studies could reveal how different types of innovation ecosystems (technology parks, industry clusters, digital platforms) influence the governance-innovation dynamics we identified, potentially providing insights into optimal ecosystem design for SME innovation support. However, researchers should carefully distinguish between bilateral governance mechanisms and ecosystem-level coordination to avoid theoretical confusion between these distinct but related phenomena.

6.2.5. Resource orchestration and dynamic capabilities

Our finding that resource orchestration capabilities may be more important than absolute resource levels for SME innovation success opens important avenues for capability-focused research. Future studies could develop more precise measures of orchestration capabilities and examine their development processes, examining how SMEs acquire, develop, and deploy these meta-capabilities over time.

Research could investigate the specific micro-processes through which SMEs combine internal and external resources for innovation, identifying the organizational routines, managerial practices, and learning mechanisms that enable effective orchestration. Understanding these processes could inform capability-building interventions and training programs for SME managers.

Dynamic capabilities research could examine how SMEs reconfigure their resource portfolios and governance approaches in response to changing innovation requirements, market conditions, and technological developments. Such research would contribute to understanding organizational adaptation and resilience in innovation contexts.

6.2.6. Intervention and policy research

Intervention studies examining the effectiveness of specific policies, programs, or management practices designed to enhance SME innovation capabilities through improved governance would provide valuable practical insights while contributing to theory development. Such studies could test the real-world effectiveness of our recommendations while generating new understanding of implementation challenges and success factors.

Policy experiments could compare different approaches to SME innovation support, contrasting resource provision with capability building, individual firm support with ecosystem development, and direct funding with governance facilitation. Such research could inform evidence-based policy design and resource allocation decisions.

Training program evaluations could examine the effectiveness of interventions designed to help SME managers develop resource orchestration capabilities, governance configuration skills, and partnership management expertise. Understanding what works in practice would enhance the practical utility of academic research.

6.2.7. Cultural and institutional context research

International comparative research examining governance-innovation relationships across different cultural, institutional, and economic contexts could provide valuable insights into the universality versus context-specificity of our findings. Such research could illuminate how national business systems, cultural values, institutional frameworks, and economic development levels influence the effectiveness of different governance approaches for fostering innovation.

Research in different emerging economies could examine whether our findings from Morocco's textile industry apply to other traditional manufacturing contexts undergoing modernization pressures. Studies in developed economies could reveal whether resource orchestration capabilities remain important when absolute resource constraints are less severe.

Cultural research could examine how trust development, communication patterns, and authority relationships vary across cultures and influence governance-innovation dynamics. Understanding these cultural contingencies would enhance the practical applicability of governance theory across different contexts.

6.2.8. Methodological innovations

Future research could employ innovative methodological approaches to advance understanding of governance-innovation relationships. Mixed-methods designs combining qualitative depth with quantitative breadth could provide more comprehensive insights than either approach alone.

Digital research methods, including social network analysis of innovation partnerships, sentiment analysis of communication patterns, and machine learning approaches to pattern recognition in governance data, could reveal new insights into complex governance-innovation dynamics.

Experimental and quasi-experimental designs could provide stronger causal evidence about governance effects on innovation outcomes, though careful attention to ethical considerations and external validity would be essential.

This comprehensive research agenda reflects the complexity of governance-innovation relationships while highlighting specific directions that could advance both theoretical understanding and practical application. The diversity of suggested approaches recognizes that different research questions require different methodological strategies and that multiple perspectives are needed to fully understand these complex phenomena.

6.3. Concluding reflections

This study contributes to a growing understanding that innovation in contemporary business environments increasingly depends on the ability to orchestrate complex networks of relationships rather than simply possessing superior internal capabilities. For SMEs operating in global supply chains, this reality is particularly pronounced, as resource limitations necessitate external collaboration while power asymmetries create governance challenges that must be carefully managed to enable innovation success.

Our findings suggest that the future competitiveness of SMEs may depend less on their ability to compete independently and more on their capability to collaborate effectively within innovation ecosystems. This shift requires new theoretical frameworks that integrate governance, innovation, and network perspectives while providing practical guidance for managing the inherent tensions between collaboration and competition, openness and protection, trust and control.

As global supply chains continue to evolve and innovation requirements intensify, understanding and managing the governance-innovation nexus will become increasingly critical for SME survival and success. Our study provides a foundation for this understanding while highlighting the need for continued research and practice development in this vital area of organizational and economic activity. The journey toward fully understanding these complex relationships has only begun, but the insights generated here provide valuable stepping stones for future exploration and practical application in the challenging but rewarding world of SME innovation and governance.

CRediT authorship contribution statement

Hanane Bougadir: Writing – original draft, Resources, Investigation, Conceptualization. **Antonio García Sánchez:** Writing – review & editing, Supervision, Conceptualization. **Mustapha Zahir:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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